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ABBREVIATIONS USED IN THE PAPER

AI – Artificial Intelligence

BERT – Bidirectional Encoder Representations from Transformers

DL – Deep Learning

DT – Decision Tree

EDA – Exploratory Data Analysis

EU – European Union

FN – False Negative

FP – False Positive

GCN – Graph Convolutional Network

GNN – Graph Neural Network

LIME – Local Interpretable Model-agnostic Explanations

LR – Logistic Regression

LSTM – Long Short-Term Memory

ML – Machine Learning

MLMA – Multilingual and Multi-aspect

NB – Naive Bayes

NLP – Natural Language Processing

PRISMA – Preferred Reporting Items for Systematic Reviews and Meta-Analyses

RF – Random Forest

RNN – Recurrent Neural Network

RoBERTa – Robustly Optimized BERT Pretraining Approach

SHAP – SHapley Additive exPlanations

STD – Standard deviation

SVM – Support Vector Machine

TN – True Negative

TP – True Positive

XAI – Explainable Artificial Intelligence

INTRODUCTION

Article 86 of the EU Artificial Intelligence Act (European Union, 2024) establishes the "Right to Explanation of Individual Decision-Making." This provision grants individuals the right to obtain clear and meaningful explanations when decisions significantly affecting their health, safety, or fundamental rights are made based on outputs from high-risk AI systems. This legislative development underscores the growing recognition that explainability is not merely a technical consideration but a fundamental requirement for responsible AI deployment.

Social media platforms increasingly rely on AI assisted systems to detect and remove detrimental content, including hate speech, at unprecedented scale and speed (Gongane, Munot and Anuse, 2022). These systems make millions of decisions daily that directly impact user expression, platform safety, and the quality of public discourse. State-of-the-art AI models that achieve high precision for detection of detrimental content may not be able to interpretable to the users of the models.

Explainable Artificial Intelligence (XAI) has emerged as a field dedicated to addressing the interpretability challenges of complex AI systems. By providing insights into how and why AI models make specific decisions, XAI methods aim to enhance transparency, facilitate human oversight, and build trust in automated systems. In the context of hate speech detection, XAI techniques offer potential benefits for content moderators, platform users, and regulatory compliance. However, the reliability and consistency of these explanations remain an understudied aspects that directly impact their practical utility and trustworthiness.

This thesis investigates the consistency of explanations generated by two XAI methods of Local Interpretable Model-agnostic Explanations (LIME) and SHapley Additive exPlanations (SHAP) when applied to hate speech detection models for social media content. By systematically evaluating the consistency of these explanations across multiple runs, this research contributes to the understanding of XAI reliability.

The research object of this study is XAI methods applied to hate speech detection models on social media. Specifically, the investigation focuses on LIME and SHAP, two post-hoc explanation techniques that can be applied to various machine learning models, including deep learning architectures commonly used for natural language processing tasks.

The research subject is defined as the consistency of XAI methods when applied to AI models that detect hate speech. Consistency in this context refers to the consistency of feature attributions and importance rankings across multiple explanation instances for the same prediction provided from the XAI method.

This research is guided by the hypothesis that SHAP provides more consistent explanations than LIME when applied to hate speech detection models, as measured by consistency metrics. This hypothesis is grounded in theoretical differences between the two methods: while LIME relies on local approximations through random perturbations, SHAP is based on cooperative game theory principles that may provide more consistent attributions. **The primary research question asks: How consistent are the explanations generated by LIME and SHAP when applied to for hate speech detection models across multiple runs?**

To address this research question, this thesis employs a mixed-methods approach combining quantitative and qualitative techniques. The methodology incorporates several quantitative methods, beginning with dataset aggregation and preprocessing, which involves combining multiple hate speech datasets to create a corpus that captures diverse samples of hate speech across different platforms and targeted groups. Statistical analysis performed by applying consistency metrics including Jaccard Similarity, Spearman Correlation, and Kendall Tau-b to evaluate the consistency of XAI explanations across different random seeds. The quantitative approach is completed through comparative analysis, systematically comparing LIME and SHAP explanations across different classification outcomes to assess consistency patterns in various prediction scenarios. Complementing these quantitative methods, the research employs qualitative visual interpretation, analysing visualization outputs from both LIME and SHAP to assess interpretability and practical utility.

At its foundation is model selection and configuration, utilizing the pre-trained CardiffNLP model, a RoBERTa-based architecture specifically fine-tuned for hate speech detection on social media content. This is coupled with controlled experimentation, applying XAI methods with consistent parameters across multiple random seeds to isolate the effect of stochasticity on explanation consistency. The experimental approach is further refined through stratified sampling, selecting high-confidence predictions across different classification outcomes (true positives, true negatives, false positives, false negatives) to ensure evaluation across diverse prediction scenarios that represent model performance.

The thesis is structured as follows: Chapter 1 provides a comprehensive review of the literature on hate speech detection, XAI methods, and evaluation frameworks, establishing the theoretical foundation and identifying research gaps. Chapter 2 details the methodology, including dataset construction, model selection, experimental design, and evaluation metrics. Chapter 3 presents the results of both quantitative and qualitative analyses, examining consistency patterns across different XAI methods and prediction scenarios. Finally, the conclusion synthesizes the findings, discusses their implications for XAI research and content moderation practices,

acknowledges limitations, and suggests directions for future research. Through this structured investigation, the thesis aims to contribute meaningful insights to the growing body of knowledge on explainable AI and its application in addressing online hate speech.

1. STATE OF THE ART

1.1. Literature Search Strategy

Hate speech is one form of detrimental content found on social media. According to Gongane, Munot and Anuse (2022) there are many types of detrimental content such as hate speech, spam, rumours, fake news, toxic, violent, etc. This detrimental content detection is assisted with AI-based techniques with NLP using ML and DL. These techniques are considered black box models that can predict if content is detrimental, but not on why the model choses it to be detrimental. According to research (Hassija et al., 2023) XAI methods have emerged that attempt to apply post-hoc methods to black-box models to make them interpretable. This interpretability could aid human moderators in content moderation and transparency to users.

A systematic search process following PRISMA guidelines is detailed in Figure 1 to begin a literature review for detrimental content on social media and XAI use.

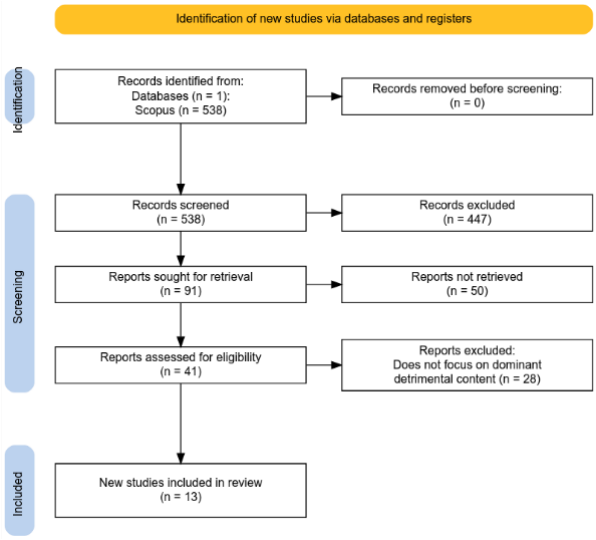

Fig. 1. PRISMA Search Strategy created with PRISMA2020 (Haddaway et al., 2022)

1.1.1. Search Database and Justification

Scopus was used as the search database due to its comprehensive and curated nature. According to Baas et al. (2020) we can determine high quality curation is indexed through continuous evaluation due to institutional profiles, constant monitoring of source data, and determination of new journal/publisher participation of the exploitative practices to get research published. This aids in the search strategy to include high quality research in our literature search.

1.1.2. Query Construction and Keyword Categories

The search query was constructed by combining keywords across several distinct categories relevant to the research focus. These categories, along with their corresponding search terms and Boolean operators, are detailed in table 1. These categories and their respective keywords were combined using the Boolean operator 'AND' to create a focused search string tailored to the Scopus database.

table 1

Search Strategy Components and Justification

Search Concept	Search Terms and Boolean Operators	Rationale for Inclusion
Social media	((social OR online) OR (user-generated OR (user AND generated)) AND (media OR network*))	Identifies papers related to social media platforms where content moderation is relevant (Saheb et al., 2024; Gongane et al., 2022).
Detrimental Content Keywords	(content AND (mod* OR detect*)) OR bias OR hate OR fake OR harassment OR spam OR toxic OR abusive	Focuses on detrimental content and detection of specific violations like hate speech, misinformation, and harassment (Gongane et al., 2022).
Explainable AI	(model-agnostic OR shap OR lime OR explaina* OR white-box OR (white AND box) OR black-box OR (black AND box) OR xai)	Targets papers discussing XAI techniques and terms for interpreting AI models used in content moderation (Elicit.com, 2024; Paperdigest.org, 2024; Molnar, 2022).
Publication Date	PUBYEAR > 2019 AND PUBYEAR < 2025	Ensures the research is recent and reflects the evolving field of XAI.
Language	LIMIT-TO (LANGUAGE, "English")	Limits results to English language papers for analysis.

1.1.3. Inclusion and Exclusion

The eligibility of each identified record was assessed based on topic focus, use of XAI techniques, platform relevance, and accessibility. A detailed justification for the inclusion or exclusion of key studies is provided in table 2. This table builds upon Table 3 by directly linking decisions to the search components defined earlier.

table 2

Inclusion and Exclusion Criteria

Criteria	Include	Exclude
Topic Focus	Studies focused on detrimental content (hate speech, fake news, toxic, abuse, harassment, spam, bias)	Studies focused on non-detrimental content topics
XAI Techniques	Uses or discusses explainable AI (XAI) (e.g., SHAP, LIME)	Does not include XAI techniques
Platform	Research applied to social media platforms (e.g., Twitter)	Not conducted on social media or platforms
Accessibility	Accessible full-text articles	Not accessible or retrievable

1.1.4. Search Results and Screening Process

Following the query execution in Scopus, a total of 538 records were identified. As outlined in the PRISMA diagram (Figure 1), all records were initially screened based on titles, leading to

the exclusion of 447 non-relevant studies. The remaining 91 reports were retrieved for full-text assessment; however, 50 of these could not be accessed and were excluded based on inaccessibility. The eligibility of the remaining 41 full-text studies was then assessed according to the defined inclusion and exclusion criteria. This process emphasized relevance to detrimental content, application of XAI techniques, and focus on social media platforms. Table 3 demonstrates a sample of the screening process using the criteria outlined in table 2.

table 3

Eligibility and Justification of Research Records

Record	Eligible	Justification
Fine-Grained Multilingual Hate Speech Detection Using Explainable AI and Transformers (Siddiqui et al.,2024)	Yes	Uses the model XLM-RoBERTa to detect hate speech on Twitter, interprets results from predictions with LIME
A comprehensive review of trustworthy, ethical, and explainable computer vision advancements in online social media (Fahim Anzum et al., 2024)	No	Not accessible
A SHAP-based controversy analysis through communities on Twitter (Samy Benslimane et al., 2024)	Yes	Uses SHAP, an XAI technique, to analyse rumours on Twitter, which is a social media platform.
A method of classification twitter posting location for a specific space (Hara et al., 2021)	No	While using XAI techniques on Twitter, the focus is on geolocation rather than detrimental content.
Explainability of Depression Detection on Social Media: From Deep Learning Models to Psychological Interpretations and Multimodality (Ana-Sabina Uban, Chulvi and Rosso, 2022)	No	Uses XAI and is on social media but is focused on depression detection.

Once the eligible 41 reports were determined, manual tagging of the reports for which detrimental content was the focus. To narrow down the scope for detrimental content detection, the dominant detrimental content tag is used. The tags were extracted using python (appendix 1) to visualize which content was the most prominent. Figure 2 shows that hate (speech) was the dominant detrimental content on social media and XAI was used for the years between 2021 and 2024. Out of the 41 eligible reports, 13 reports about hate speech were included in the following literature review.

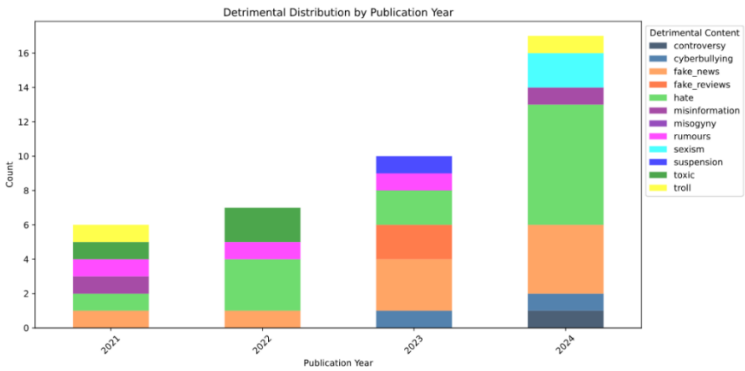

Fig. 2. Detrimental Content Distribution by Publication Year

1.2. Literature Review

Table 4 is an overview of hate speech detection studies using XAI chosen from the literature search. These studies offer insights into how XAI can enhance the transparency and understanding of models designed to identify such detrimental content. An overview of the 13 selected studies, detailing aspects such as their respective dataset sources, the specific models on which XAI techniques were applied, the XAI methods utilized, and the particular use of XAI in each case.

1.2.1. Dataset Source and Task

The common data source for hate speech is Twitter (also known as X) shown from Table 4. Lesser-known social networks as Gab and Resset appear once along with Facebook and YouTube appearing twice each. Most of the datasets used the text content of users post except for the study by Wich et al. (2021) which also used post history and social network data of the user.

Binary classification was the classification task that appeared the most with it being defined with the definition of hate speech in their respective study. Multiclass classification used the distinction between offensive and hate speech. Fine-grained classification included normal speech and hate speech that had subcategories for who the hate speech targeted.

1.2.2. Models

The column of "XAI Used on AI Model" from table 4 shows that explainable AI techniques were applied across a wide range of model types, highlighting the adaptability of XAI methods to both traditional and deep learning models. LSTM and FAST-RNN appear multiple times. Transformer-based models such as BERT, DistilBERT with GraphSAGE, and XLM-RoBERTa also feature prominently, indicating efforts to explain complex, contextual language representations. The inclusion of Support Vector Machines and Naive Bayes variants demonstrates XAI applications on classic linear classifiers. Tree-based models like RF, DT, and XGBoost are also present along with an instance of LR. Deep learning models of RNNs and transformers are the prominent models according to table 4.

table 4

Overview of Hate Speech Detection Studies Using XAI

Reference	Dataset Source	Classification Task	XAI Used on AI Model	XAI Method	XAI Use
Ansari, Kaur and Saxena (2023)	YouTube, Facebook, Twitter	Binary	LSTM	LIME, Integrated Gradient	4 examples: Local explanations using and metrics of AOPC, log-odds, and coherence
Hashmi et al. (2024)	Twitter datasets and multilingual corpora	Binary	SVM	LIME	23 examples: Two for each language, except one with 20 examples (1.00 prob.)
Wich et al. (2021)	Twitter	Binary	DistilBERT with GraphSage	SHAP	6 examples: Explanation of each sub model, 1 example with 1.00 prob. Also used masking strategies for component and feature level
Imbwaga, Chittaragi and Koolagudi (2024)	YouTube	Binary	RF	LIME	2 examples: Hate-labelled examples in English and Kiswahili
Nandini and Schmid (2023)	Twitter	Multiclass	BERT	LIME	4 examples: One hate-labelled + aggregated features from 50 examples/class
Siddiqui et al. (2024)	Twitter and cyber bullying datasets	Fine-grained	XLM-RoBERTa	LIME	9 examples: TP, TN, FP, FN + 5 fine-grained samples, 4 with 1.0 prob
Tiwari (2024)	Twitter	Binary	LSTM	LIME	4 examples: One for each LSTM word embedding method
Yadav, Kaushik and McDaid (2023)	Twitter	Binary	NB-B	LIME	4 examples: One TP and one FP, with one at 1.0 prob
Babaeianjelodar et al. (2022)	Twitter	Multiclass	XGBoost	SHAP	1 example: Explained hate-labelled tweet
Hashmi and Sule Yildirim Yayilgan (2024)	Facebook, Twitter, Resset	Multiclass	FAST-RNN	LIME	12 examples: Two per class + two misclassified
Mehta and Passi (2022)	Twitter, Gab, Wikipedia	Binary	RF, NB-G, DT, LR	LIME	4 examples: DT, RF, LR, NB models. IOU F1, Token F1, AUPRC, Comprehensiveness, Subgroup AUC, etc.
Mazhar Qureshi, Qureshi and Rashwan (2023)	Twitter	Multiclass	SVC	LIME, SHAP, Counterfactual	21 examples: SHAP/LIME examples across classes + counterfactual text edits
Hareem Kibriya et al. (2024)	Twitter	Fine-grained	LSTM	LIME, SHAP	7 examples: 4 LIME (weighted features), 2 SHAP-based instances

1.2.3. XAI Methods and Use

Table 4 displays the most used XAI method is LIME followed by SHAP with one instance of integrated gradients and one instance of counterfactuals. The selection of the XAI is justified by many of the studies due to its ability to interpretant a couple of instances and is applicable to any model.

All studies in the literature review illustrated selected examples of text to apply their selected XAI method and interpret the explanation provided. Figure 3 illustrates the general process for using XAI to provide explanations for decisions made by a ML/DL model to detect hate speech. The selected instances of text were chosen in inconsistent methods. A common selection method shown in the studies was selecting examples where the respective AI model made TP, TN, FP, and/or FN predictions from classification task and having a prediction probability of 1.00 in the model's decision.

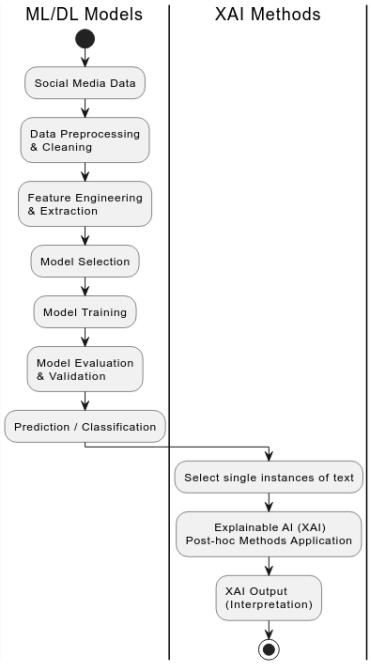

Fig. 3. Process for Using XAI Methods on ML/DL models from research from Gongane, Munot and Anuse (2024)

5 studies used XAI in an additional method as previously stated. The study conducted by Ansari, Kaur and Saxena (2023) compared 2 pairs of explanations and used quantitative metrics to compare explanations provided by LIME and integrated gradients. Wich et al. (2021) research used masking strategies for component and feature level for the multimodal classification model since it used text, history, and network of the user. Research from Mehta and Passi (2022) used a variety of ML algorithms then applied LIME followed by using explainability metrics to measure against the ERASER benchmark (DeYoung et al., 2019) which provides human rationales for NLP tasks. Nandini and Schmid (2023) research also aggregated features provided by LIME to obtained

frequent words obtained from the different classes of Hate, Offensive and None. Mazhar Qureshi, Qureshi and Rashwan (2023) research proposed the use of LIME to provide counterfactuals to neutralize hate speech.

1.2.3. Identified Gaps from Literature Review

Despite the increasing application of LIME and SHAP, there remains a gap in standardized methodologies for evaluating the consistency, practical utility and actionability of their explanations in real-world content moderation workflows. Few studies have explored in the literature review explore the comparative effectiveness of different XAI techniques for hate speech.

1.3. State of the Art

This section delves into the current state of the art concerning hate speech, its detection using AI and the role of XAI in understanding these complex systems.

1.3.1. Introduction to Hate Speech in Social Media

For this thesis, we will use the definition provided by the United Nations (2024) as "any kind of communication in speech, writing or behaviour, that attacks or uses pejorative or discriminatory language with reference to a person or a group on the basis of who they are, in other words, based on their religion, ethnicity, nationality, race, colour, descent, gender or other identity factor. " Rather than focusing solely on legalistic interpretations or narrow academic constructs, this definition allows for an exploration of hate speech as a phenomenon that erodes social cohesion and incites discrimination. It is to be noted that the act of defining hate speech can be a multifaceted occurrence as noted by Rawat, Kumar and Surender Singh Samant (2024) with a variety of social media platforms and organizations defining hate speech in a variety of ways that include or exclude different factors.

The pervasiveness of detrimental content on the internet, particularly on social media platforms, has become a societal concern with far reaching implication for countries worldwide. As highlighted by Kiritchenko, Nejadgholi, and Fraser (2021) in their survey on confronting abusive language, exposure to toxic and hateful online comments can lead to severe psychological trauma for individuals, contribute to radicalization of others, and possibly incite self-harm and/or violence. The victims of such online abuse are often the most vulnerable segments of society, including minorities, the LGBTQ+ community, and women. Leading victims to their

marginalization and silencing, which in turn diminishes the diversity of viewpoints which are essential for a healthy national discourse (Kiritchenko, Nejadgholi, and Fraser, 2021).

1.3.2. Hate Speech Detection using AI

The landscape of AI-driven hate speech detection has been systematically analysed in recent literature, such as the work by Rawat, Kumar, and Samant (2024), which explored research questions covering recent efforts, common topics, data sources, techniques, performance metrics, and available dataset in this domain.

The detection of hate speech using AI aims to address the proliferation of harmful content on social media and other online platforms. Recent advancements span various methodologies, leveraging ML, DL, ensemble models, GNN, GCN to identify and classify hate speech as shown in figure 4. DL models are particularly effective for analysing vast and complex textual datasets. The DL technique which is predominantly to detect hate speech is BERT followed by LSTM as shown in figure 5.

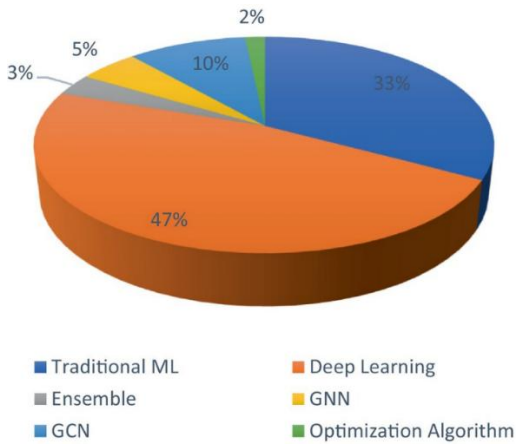

Fig. 4. Percentage distribution of various AI techniques in hate speech detection survey by Rawat, Kumar and Surender Singh Samant (2024)

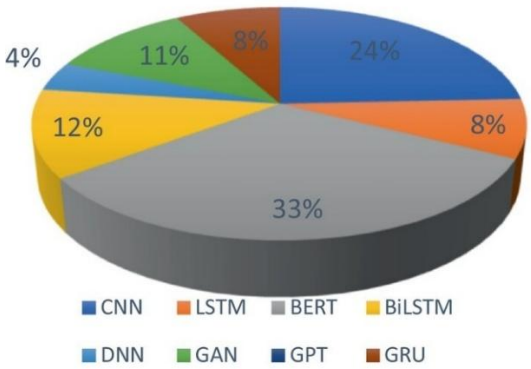

Fig. 5. Percentage distribution of various DL techniques in hate speech detection survey by Rawat, Kumar and Surender Singh Samant (2024)

1.3.3. Explainability in Artificial Intelligence (XAI)

With the use of ML/DL models, these are models seen as black box models which are able provide predictions but are not able to give a reason for these predictions. XAI provide methods to be able to provide these black box models an explanation to their decision. Barredo Arrieta et al. (2020) states one hindrance is that the misuse of interpretability and explainability in literature. Interpretability is a model's inherent characteristic, indicating how easily a human can understand its inner workings. In contrast, explainability is an active feature, referring to any action or process taken to make a model's operations clear.

Research by Ali et al. (2023) provides a knowledge graph in Figure 6 how concepts such as interpretability and explainability assist in trustworthiness with the use of XAI. This trustworthiness aids the use of AI in a variety of fields and can satisfy requirements of developers, researchers, regulators and other stakeholders.

Fig. 6. Relations among XAI concepts from Ali et al. (2023)

The XAI methods LIME and SHAP are only two model-agnostic techniques that can be applied after predictions (post-hoc) are made. Barredo Arrieta et al. (2020) splits XAI methods of ML into models that are transparent such as linear/logistic regression, post-hoc explainability which can be specific to a model or agnostic. Model-agnostic are further segmented with more subsections as well as model specific methods.

1.3.4. Evaluation Metrics of XAI

A critical aspect of XAI evaluation that warrants particular attention is the prevalent use of anecdotal evidence in assessing explanation quality. According to a comprehensive systematic review by Nauta et al. (2023), one in three papers out of 300 papers surveyed introducing XAI methods evaluate exclusively with anecdotal evidence, relying on individual, convincing

examples. This approach to evaluation is often based solely on the researcher's intuition of what constitutes a good explanation rather than objective, quantifiable metrics.

Evaluating the quality and effectiveness of explanations generated by XAI methods is required to assess their effectiveness. As detailed by Schwalbe and Finzel (2023) in their comprehensive taxonomy, metrics for XAI can be broadly categorized based on the level of human involvement required for their measurement. These categories are visualized in figure 7 which are functionally grounded, human grounded, and application grounded metrics. Each category provides a different metric through which to assess the performance and utility of an explanation.

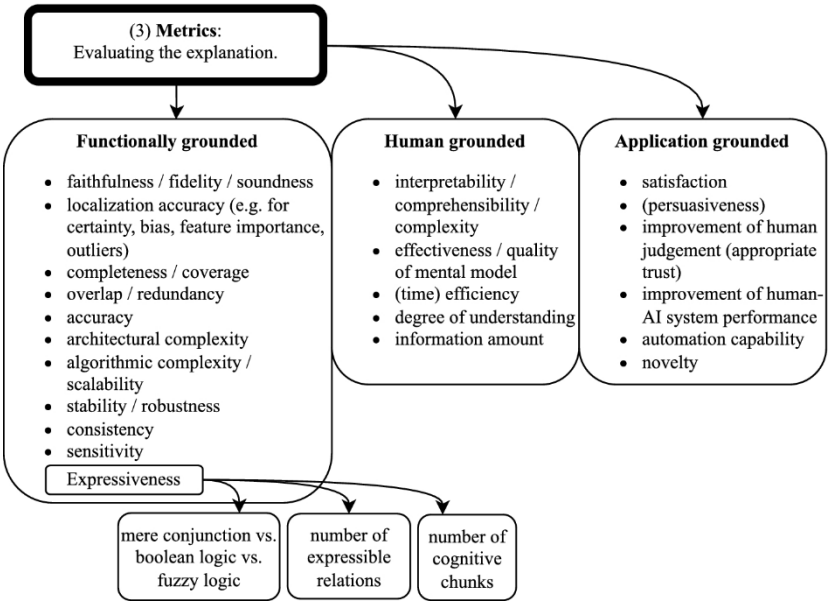

Fig. 7. Overview of XAI metrics by Schwalbe and Finzel (2023)

Schwalbe and Finzel (2023) goes on to define functionally grounded metrics assess the formal properties of an explanation method itself, without necessitating direct human feedback. Metrics for functionally grounded with metrics such as accuracy, sensitivity, and consistency. Human grounded metrics, in contrast, directly involve humans in the evaluation loop. These metrics typically rely on tasks specifically designed to assess how well a person can understand, interpret, or utilize the explanations provided by an XAI method with metrics such as information amount and interpretability. Finally, application-grounded metrics evaluate the impact and usefulness of XAI explanations within the context of the final application or task the AI system is designed for using metrics such as satisfaction and improvement of human judgement.

1.3.5. Challenges and Future Research Directions

A primary area for future research involves a deeper exploration of the effects of hate speech and the complexities surrounding it definition. The very definition of hate speech remains a contentious issue, varying across platforms, legal jurisdictions, and cultural norms. Research could explore the development of more universally applicable yet contextually sensitive frameworks for defining and categorizing hate speech.

Future research could focus on developing more robust, context-aware, and adaptable AI models. This includes exploring hybrid approaches that combine linguistic features with deep learning, investigating multimodal detection that incorporates images or audio, and addressing the significant challenge of bias in training data, which can lead to discriminatory model behaviour. Concurrently, as discussed in Section 1.3.3, the application of XAI methods like LIME and SHAP to these complex models is essential for transparency and trust though there are other XAI methods that are usable. Future research could investigate the limitations of current XAI techniques when applied to hate speech detection, explore new XAI methods tailored for natural language processing tasks, and examine how explanations can be made more meaningful and actionable for end-users, including content moderators and policymakers.

As outlined in Section 1.3.4, evaluating the quality of explanations is a complex task, with existing metrics categorized as functionally grounded, human-grounded, and application-grounded. Focus on refining existing metrics and proposing new ones that are specifically suited to the complexities of hate speech explanations is a possible research direction. This includes developing benchmarks for comparing different XAI methods and their outputs, investigating the correlation between different types of metrics, and understanding how human evaluators perceive and utilize explanations in real-world scenarios.

2. DATA & METHODOLOGY

2.1. Dataset Aggregation and Preprocessing

The initial phase of this research involved the aggregation and preprocessing of multiple existing datasets to construct a unified corpus for the subsequent hate speech detection and explainability analysis. This section outlines the chosen datasets and the rationale behind their selection, details of data processing and merging procedures undertaken to unify formats and provides an overview of the final structured dataset that forms the dataset used in this study. GitHub repository with all data and code for this thesis is available to be accessed in appendix 1.

2.1.1. Source Datasets and Rationale for Selection

HateXplain (Mathew et al., 2021), MLMA (Ousidhoum et al., 2019), and Measuring Hate Speech (Kennedy et al., 2020) are the datasets we choose to utilize. Each dataset is used for hate speech detection and had annotated data for not only hate speech, but also the targeted group the hate speech is directed at. The datasets are sourced from Twitter which was the dominant source of data (Section 1.2.1) along with other social media sites of YouTube, Reddit and Gab.

2.1.2. Data Processing and Merging Process

Table 5 displays the constructed columns of the created dataset. Based on our definition of hate speech (Section 1.3.1) `label_hatespeech_binary_offensive_included` is used in our experiments. Our dataset is available (appendix 1) to be used with different definitions of hate speech and hate speech classification tasks.

table 5

Descriptions of Columns in the Hate Speech Dataset

Column Name	**Description**
id_new	Created
text	Unaltered original text
extracted_target	Extracted from original targets of hate speech
label_hatespeech.binary.offensive.not_included	Offensive not considered hate speech (binary)
label_hatespeech.binary.offensive.included	Offensive included as hate speech (binary)
label.normal.offensive.hatespeech	Normal, offensive, or hate speech (multi-class)
platform	Source of text
original.dataset.title	Original dataset name
original_id	Original dataset ID
original_label	Original label from dataset
original_target	Majority-annotated targets of hatespeech

To create a dataset for hate speech analysis, a systematic data merging process was implemented, beginning with the preprocessing of the three selected datasets. This phase was due

to inherent variations in data annotation formats and structures across the datasets. Preprocessing steps involved the standardization of columns, such as the creation of `original_id` to retain source-specific identifiers and `platform` to denote the social media origin. Furthermore, varied hate speech labels were aligned through the generation of consistent label columns, including binary and multiclass labels to distinguish among normal, offensive, and hate speech categories. For the "Measuring Hate Speech" dataset, which utilized a continuous hate speech score, predefined thresholds defined from Kennedy et al. (2020) were applied to map theses scores to the standardized categorical labels. The annotation of target groups also required normalization; for instance, HateXplain employed majority voting for target community identification, while MLMA and "Measuring Hate Speech" had their own structured approaches to categorizing targeted groups. Platform information was explicitly extracted or mapped from existing data fields, ensuring consistency across the combined dataset.

Following dataset preprocessing, the data merging process began with the concatenation of the processed dataframes from HateXplain, MLMA, and "Measuring Hate Speech" into a singular dataframe. Post-concatenation, the merged dataset underwent further refinement, including the removal of entries with ambiguous platform information to maintain data integrity. Another step in the merging procedure was the simplification and categorization of the detailed `original_target` attributes into broader identity factors which were grouped in "Measuring Hate Speech" – race, religion, nationality, gender, sexuality, disability, and other- based on a review of all unique target labels in the combined data. This resulted in the creation of a new `extracted_target` column to store the standardized, set-based classifications, with entries lacking a specific target labelled as `none`.

To ensure the uniqueness of records, duplicate entries were identified and removed based on the textual content of the posts. Finally, a new unique identifier, `id_new`, was assigned to each record in the consolidated dataset by resetting the index.

2.2. Exploratory Data Analysis of the Hate Speech Dataset

Following the dataset preprocessing, an EDA (appendix 1) was conducted to gain insights into the fundamental characteristics and distributions within the created hate speech dataset. This analysis is for understanding the composition of the data that will be used for model evaluation. The EDA focuses on the distribution of class, the source of data across different social media platforms, and the distribution of various target groups identified in the dataset.

2.2.1. Distribution of Labels and Platform

To understand the balance of classes within the dataset, the distribution of the primary label used for hate speech detection, `labe_hatespeech_binary_offensive_included`, was examined. As illustrated in Figure 8, the dataset contains a substantial number of instances labelled as normal as well as those classified as `hatespeech/offensive`. Appendix 5 contains the distribution of hate speech labels not used in our experiment from table 5. Depending on the definition of hate speech to include offensive speech as hate speech or as its own classification, label distribution is altered.

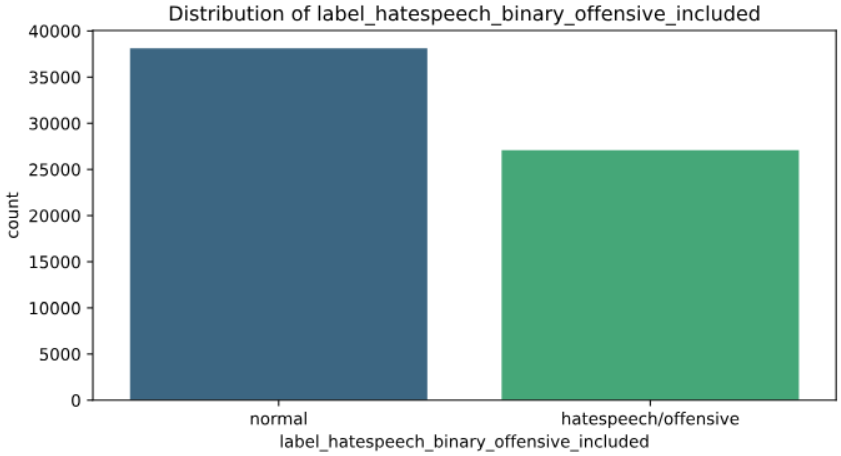

Fig. 8. Distribution of Hate speech

The dataset uses text from various online platforms. Figure 9 presents the distribution of data according to its source platform. The visualization indicates that Twitter is the most represented platform, followed by Reddit, Gab, and then YouTube.

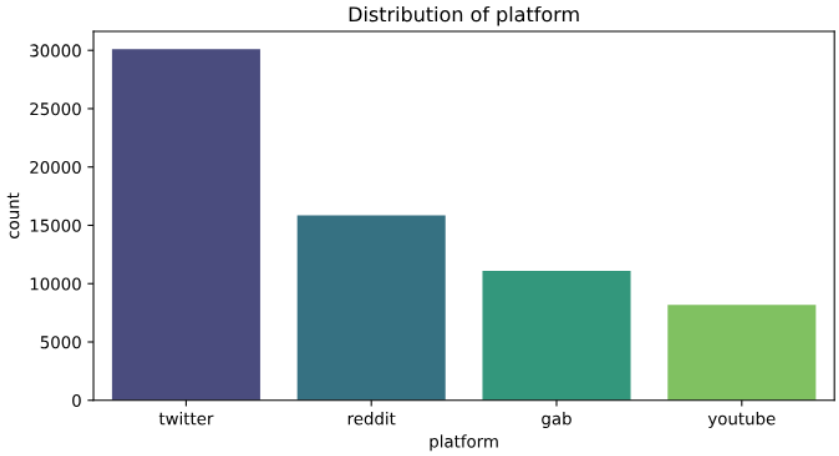

Fig. 9. Distribution of Social Media Platforms

2.2.3. Analysis of Target Groups

An aspect of this dataset is the annotation of targeted groups since hate speech is directed at a targeted group from our definition which allows for a more nuanced understanding of the nature of the hateful content. The analysis of these target groups was approached in two ways. The target

group is in column `extracted_target` which is possible to have multiple targets, one target, or none.

Figure 10 illustrates the count of each identified target group where multiple targets could be associated with a single text instance. This shows that 'gender' and 'race' are the most frequently identified target categories when multiple attributions are allowed, followed by 'nationality', 'religion', and 'sexuality'. The categories 'none', 'other', and 'disability' are also present but represent a smaller proportion of the instances.

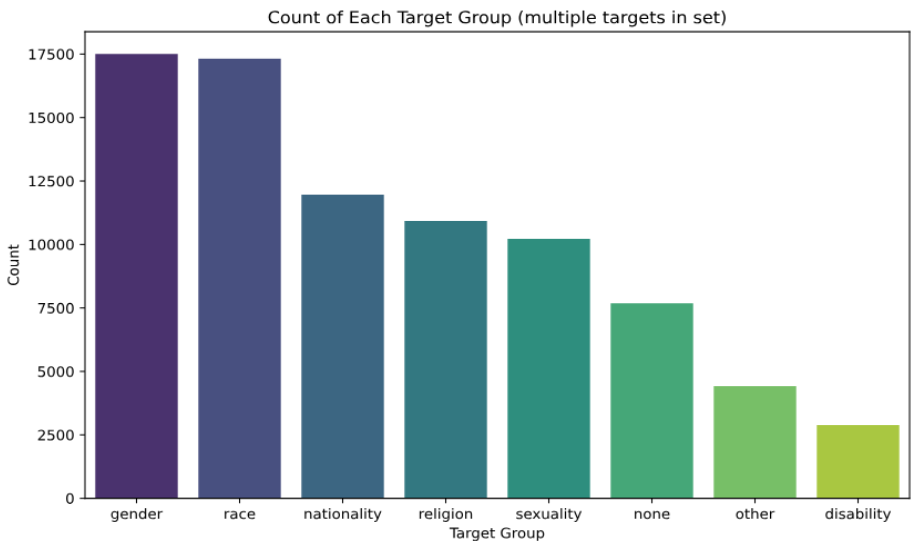

Fig. 10. Count of Each Target Group with multiple targets in set

Figure 11 displays the distribution when considering instances where only a single primary target group is assigned. In this representation, 'gender' remains the most frequent single target, followed by 'race'. Interestingly, 'none' (indicating no specific target group was identified as primary) becomes more prominent in this view compared to its representation in the multiple-target analysis. 'Religion' and 'sexuality' also feature significantly, with 'nationality', 'disability', and 'other' appearing less frequently as the sole primary target.

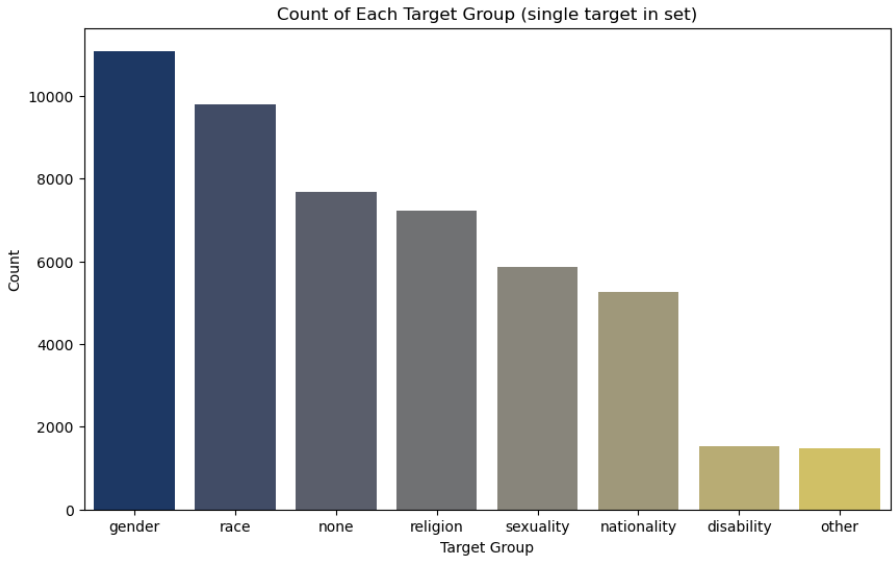

Fig. 11. Count of Each Target Group with single target in set

2.3. Hate Speech Detection Model and Setup

This section details the components of the experimental methodology employed in this thesis, focusing on the chosen hate speech detection model and the setup for its application and subsequent XAI methods.

2.3.1. The CardiffNLP Model

For this thesis, which aims to evaluate the consistency of XAI methods in the context of hate speech detection. After careful consideration of available models, the CardiffNLP model was selected. This model, developed by the Cardiff NLP group at Cardiff University, is a deep learning model based on the RoBERTa architecture and has been specifically fine-tuned for hate speech detection tasks on social media data (Antypas & Camacho-Collados, 2023)

The selection of CardiffNLP Model was driven by several key factors. Firstly, its foundation on deep learning techniques, specifically the Transformer-based RoBERTa architecture, aligns with the state-of-the-art in NLP (Section 1.3.2.) for complex language understanding tasks such as hate speech detection. RoBERTa is known for its improved pretraining methodology over BERT, leading to better performance on downstream tasks (Liu et al., 2019). Secondly, the model is pre-trained and readily available for use as cardiffnlp/twitter-roberta-base-sentiment-latest via the Hugging Face Model Hub (Hugging Face, 2025), which significantly simplifies the setup and integration process for experimental work. This accessibility allows for a focus on the XAI evaluation aspect of the research rather than on developing a hate speech model from scratch. Furthermore, the model has demonstrated strong performance, being fine-tuned on the TweetEval

benchmark, a comprehensive evaluation framework for Twitter-specific NLP tasks (Barbieri et al., 2020)

The model facebook/roberta-hate-speech-dynabench-r4-target was also considered since it has the most downloads on Hugging Face (Hugging Face, 2025) for hate speech detection. Both models were tested to determine the most suitable candidate for this research. The performance of the CardiffNLP model and the Facebook model on the test dataset is summarized in their respective confusion matrices, presented in Figure 12 and Figure 13. Given the objectives of this thesis, which focus on evaluating the consistency of XAI methods, the CardiffNLP model was ultimately selected. The higher precision of the CardiffNLP model is particularly advantageous, as it ensures that the instances identified as hate speech (and subsequently explained by XAI methods) are more reliably actual hate speech.

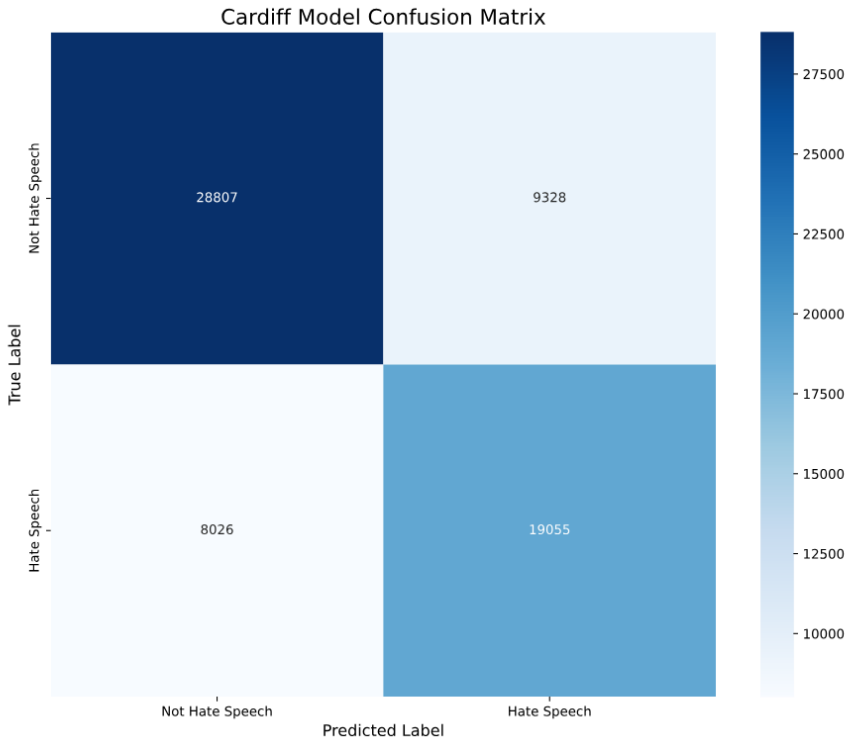

Fig. 12. Confusion Matrix for the CardiffNLP model on dataset

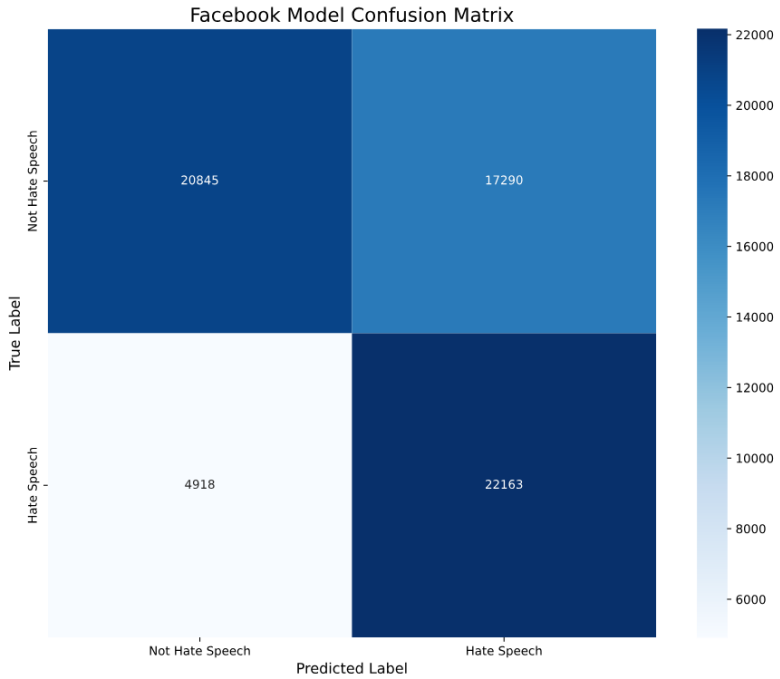

Fig. 13. Confusion Matrix for the Facebook model on dataset

The CardiffNLP model, as detailed by Antypas and Camacho-Collados (2023), was developed as part of a large-scale cross-dataset empirical evaluation aimed at creating robust hate speech detection systems. The authors highlight the challenge of dataset-specific biases in hate speech detection and propose that combining multiple datasets can lead to more generalizable models.

2.3.2. Experimental Environment

Setting up the CardiffNLP model for experimental use is straightforward due to its availability on the Hugging Face platform. The transformers library, also developed by Hugging Face (Hugging Face, 2025), provides convenient tools to download, load, and utilize the model and its associated tokenizer. The tokenizer preprocesses input text into a format suitable for RoBERTa, including tokenization into sub word units, adding special tokens, and creating attention masks. The model then processes these inputs to output a classification, typically indicating whether the given text contains hate speech or not.

The specific experimental settings employed in this study are detailed in Table 6 and available code available in appendix 1. The experiments were conducted as single-factor experiments, focusing on the consistent application of XAI methods. The computational platform utilized was Google Colab, leveraging its T4 GPU capabilities to ensure efficient model inference and XAI method computation. To ensure reproducibility and account for potential stochasticity in certain processes, a set of 4 random seeds between 0 and 100000 (838111, 14593, 3279, 97197)

were used for random number generation across libraries such as random, NumPy, PyTorch, and CUDA.

The chosen model for the experiments was the cardiffnlp/twitter-roberta-base-hate-latest version, and predictions were obtained using the Hugging Face `transformers.pipeline` configured for text classification. A batch size of 512 was employed during the prediction phase to optimize computational throughput. Several additional python packages were installed to facilitate the application and analysis of XAI methods; these included `lime` for LIME and `shap` for SHAP.

To manage computational resources and ensure a detailed analysis of high-confidence predictions, the study concentrated on a subset of the data. Specifically, 1% of the dataset instances where the CardiffNLP model assigned a prediction score of 0.95 or higher were selected. Since the selection method of single instance of instance text was used in table 4, high-confidence subset was then further divided into four categories based on the model's performance: True Positives (correctly identified hate speech), True Negatives (correctly identified non-hate speech), False Positives (incorrectly identified as hate speech), and False Negatives (incorrectly identified as not hate speech). A sample size of 163 instances was selected from each of these four categories for the subsequent XAI analysis. This stratified sampling approach ensures that the XAI methods are evaluated across a diverse range of model prediction outcomes, providing an understanding of their consistency.

table 6

Experimental Settings

Setting	**Value / Description**
Experiment Type	Single factor experiment
Platform	Google Colab T4 GPUs
Seeds	4 random seeds between 0 and 100000 [838111, 14593, 3279, 97197] (for random, numpy, torch, CUDA)
Model	CardiffNLP (cardiffnlp/twitter-roberta-base-hate-latest)
Prediction Method	HuggingFace transformers.pipeline (text-classification)
Batch Size	512
Additional Pip Packages Installed	lime, shap, hf_xet
Dataset Focus	1% of the dataset and with cardiff.score ≥ 0.95, divided into quarters of True Positives, True Negatives, False Positives, and False Negatives. Sample size per category: 163. Total size: 652

2.3.3. Experimental Process

The overall experimental process undertaken in this research is systematically illustrated in Figure 14. This process shows the workflow from data acquisition to the evaluation of XAI methods, encompassing both the machine learning/deep learning (ML/DL) model pipeline and the subsequent application and assessment of explainability techniques.

The left-hand side of Figure 14 depicts the ML/DL model pipeline. This begins with acquiring and preprocessing social media data. Subsequent steps include model selection (in this case, the pre-trained CardiffNLP model as detailed in Section 2.3.1), followed by model evaluation & validation using appropriate metrics. The pipeline culminates in prediction/classification, where the model assigns labels to new text instances. Since a pre-trained model is used, the typical stages of extensive feature engineering and model training from scratch are bypassed, with the focus being on the model's application and evaluation in its fine-tuned state.

The right-hand side of Figure 14 details the XAI methods pipeline, which builds upon the outputs of the ML/DL model. The first step here is to select single instances of text from the classified data. As described in the experimental settings, specific instances from True Positives, True Negatives, False Positives, and False Negatives are chosen for in-depth analysis. Once these instances are selected, XAI post-hoc methods application occurs. This involves applying various XAI techniques to the model's predictions for these selected instances to generate explanations for why the model made a particular classification.

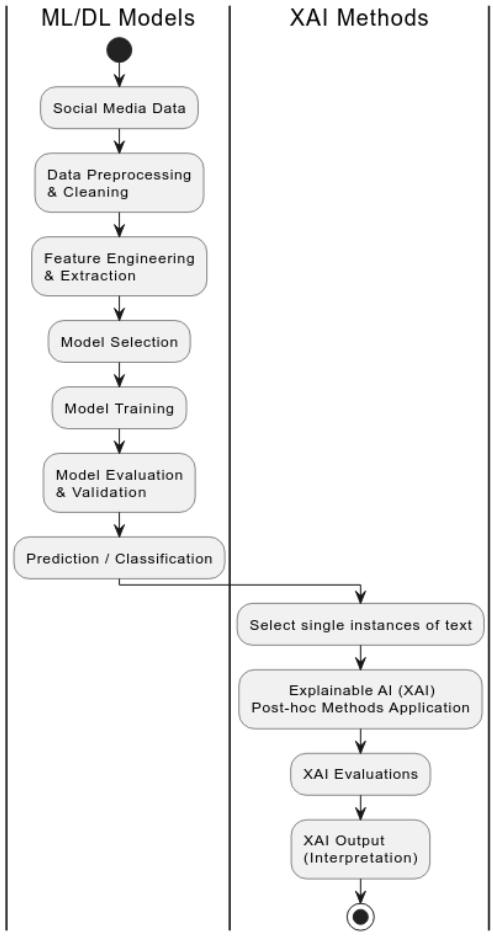

Fig. 14. Process for Using XAI methods and evaluating on ML/DL models adapted from figure 3

Following the generation of explanations, XAI Evaluations are performed. This is the stage where the quality of the generated explanations is assessed. Finally, the core focus of this thesis, the calculation and analysis of consistency metrics, is undertaken. This step evaluates how consistent the explanations provided by different XAI methods are, both with themselves across different runs for the same model and input instances.

2.4. Explainable AI (XAI) Methods

Having established the hate speech detection model and the experimental environment in the preceding sections, this section explains the core XAI methods employed in this thesis: LIME and SHAP. The primary objective of incorporating these methods is to generate explanations for the predictions made by the chosen hate speech detection model, thereby facilitating an analysis of their consistency for single instances of text.

2.4.1. LIME (Local Interpretable Model-agnostic Explanations)

LIME is a technique designed to explain the predictions of any machine learning model in a locally faithful manner (Ribeiro, Singh, & Guestrin, 2016). LIME operates by approximating the behaviour of a complex, black-box model with a simpler, interpretable model in the local vicinity of a specific instance being explained. Allowing for an understanding of which features contributed most significantly to a particular prediction for that instance.

For text classification tasks, LIME generates explanations by creating a set of perturbed samples around the instance of interest. These perturbations typically involve removing or altering words from the original text. The black-box model then makes predictions on these perturbed samples. LIME subsequently fits a sparse linear model to these predictions, and the coefficients of this linear model are then used as the feature importances, indicating how much each word (or feature) contributed to the model's decision for that specific text instance. The 'LimeTextExplainer' class from the LIME library (Ribeiro , 2021) is specifically designed for this purpose.

The explanation ξ(x) for an instance x is obtained by the following formula (1) (Ribeiro, Singh, & Guestrin, 2016); $\arg\min_{g \in G} \mathcal{L}$ denotes finding the model g from the class of interpretable models G that minimizes the following expression, $L(f, g, \pi_x)$ is the measure of how unfaithful the interpretable model g is in approximating the original model f in the locality defined by π_x, f is the original complex model being explained, g is an interpretable model from the class of interpretable models G, π_x is the locality weight function around instance x, determining the size

of the neighbourhood around x to be considered, Ω(g) is a measure of complexity of the explanation model g :

$$\xi(x) = \arg\min_{g \in G} \mathcal{L}\,(f, g, \pi_x) + \Omega(g) \tag{1}$$

2.4.2. SHAP (SHapley Additive exPlanations)

SHAP is another model-agnostic XAI technique that uses mathematical reasoning from cooperative game theory to explain the output of machine learning models (Lundberg & Lee, 2017). SHAP values provide a measure of feature importance by attributing the prediction outcome to each feature based on its marginal contribution across all possible feature coalitions. This method ensures properties like local accuracy and consistency.

For text classification, SHAP can be applied using various explainer types depending on the model. The `shap.Explainer` (Lundberg, 2018) is a general interface that can often automatically select an appropriate explainer or allow for explicit specification. For models a common approach involves using an explainer that can handle text data, often by working with token-level importance or by masking parts of the input text to simulate feature absence. The SHAP values indicate the extent to which each token or feature pushes the model's output away from a baseline (expected) value towards the actual prediction.

The SHAP explanation model is an additive feature attribution method where the explanation for a prediction has the form (Lundberg & Lee, 2017) in formula (2); g(z') is the explanation model, M is the total number of simplified input features, ϕ_0 is the base value, representing the expected model output when all simplified inputs are 0, ϕ_j is the attribution value or SHAP value for feature j by indicating its contribution to the prediction, z_j' is the binary indicator (0 or 1) for the presence of feature i in the simplified input:

$$g(z') = \phi_0 + \sum_{j=1}^{M} \phi_j z_j' \tag{2}$$

The Shapley value for a specific feature j is calculated as in formula (3); ϕ_j is the Shapley value for feature j, F is the set of all features, S is a subset of features not containing j, $|S|$ is the cardinality of set S, $|F|$ is the total number of features, $f_x(S \cup \{j\})$ is the prediction when feature j is added to set S, $f_x(S)$ is the prediction for instance x when only features in set S are present:

$$\phi_j(f, x) = \sum_{S \subseteq F \setminus \{j\}} \frac{|S|!(|F|-|S|-1)!}{|F|!} [f_x(S \cup \{j\}) - f_x(S)] \tag{3}$$

2.4.3. Experimental Application of LIME and SHAP

The practical application of LIME in this study involved generating local explanations for a curated set of text instances derived from the hate speech dataset (the structure of which is partially

described by Table 6, with experimental sampling detailed in Section 2.3.2). As outlined in the experimental settings, 163 single text instances were selected from each of the four prediction categories: TP, TN, FP, and FN, based on the CardiffNLP model's high-confidence predictions (score ≥ 0.95). This stratified sampling ensures that LIME's behaviour is assessed across various model decision outcomes.

Similar to LIME, SHAP were applied to the same curated set of 163 text instances from each of the four prediction categories (TP, TN, FP, FN) derived from the high-confidence predictions of the CardiffNLP model. The objective was to generate feature attribution scores (SHAP values) for each token in the input texts from single text instances, providing an alternative perspective on which parts of the text influenced the model's decision.

In the application of both LIME and SHAP for generating explanations, a principle of minimal intervention in parameter settings was adopted to primarily leverage the default configurations established by the respective library developers, ensuring a baseline application of these XAI methods. A comprehensive list of these default settings, alongside the explicitly set parameters, is provided in Table 7 for completeness and to aid replicability, though a detailed discussion of each default parameter is beyond the scope of this section, as the focus was on a standard application.

table 7

LIME and SHAP Parameters

XAI Method	Type / Setting	Value
LIME	Explainer Type	`LimeTextExplainer`
	random_state	Same as Experimental Settings seeds (table 6) [838111, 14593, 3279, 97197]
	Other Settings	Default values used (see details below)
	kernel_width	`25`
	kernel	`None`
	verbose	`False`
	class_names	`None`
	feature_selection	`'auto'`
	split_expression	`r'\W+'`
	bow	`True`
	mask_string	`None`
SHAP	Explainer Type	`shap.Explainer`
	Masker	`shap.maskers.Text` with corresponding model tokenizer
	seed	Same as Experimental Settings seeds (table 6) [838111, 14593, 3279, 97197]
	Other Settings	Default values used (see details below)
	masker	`None`
	link	`shap.links.identity`
	algorithm	`'auto'`
	output_names	`None`
	feature_names	`None`
	linearize_link	`True`

2.5. Evaluation Methodology for XAI Consistency

The evaluation of XAI methods extends beyond their ability to generate explanations; it involves assessing the quality of these explanations. Within this research, a key aspect of this evaluation is the concept of consistency.

2.5.1. Defining Consistency in XAI for Hate Speech Detection

Nauta et al. (2023) provide a comprehensive framework for evaluating XAI methods, in which consistency is identified as one of twelve core properties for assessing explanation quality. They define consistency as a property that "describes how deterministic and implementation-invariant the explanation method is" (Nauta et al., 2023, p. 11). This definition encompasses two key aspects: (1) determinism, which addresses whether an explanation method produces identical explanations for identical inputs, and (2) implementation invariance, which concerns whether two models that produce the same outputs for all inputs also yield the same explanations, regardless of their internal implementation.

In the context of XAI for hate speech detection, consistency refers to the change in explanations provided by an XAI technique when subjected to variations in the explanation generation process or the input data, assuming the underlying model and its prediction remain unchanged. For instance, if an XAI method produces different sets of important features for the exact same input instance and model prediction, merely due to a change in its internal random initialization (e.g., a different random seed), its explanations might be perceived as arbitrary or unreliable.

2.5.2. Consistency Metrics

Since the explanations of LIME use the coefficients of a linear model and SHAP uses shapely values for their respective feature importance, direct value comparison is not possible. They both contain features that can identified and ordered by the values they provide. To quantitatively assess the consistency of explanations generated by LIME and SHAP, three metrics were selected, each capturing a different aspect of similarity between sets or ranked lists of features. These metrics are Jaccard Similarity, Spearman Rank Correlation, and Kendall's Tau-b Rank Correlation. The experimental analysis applies these metrics to compare the top 5 features identified by LIME and SHAP across different random seed initializations for the same input instance.

2.5.3. Jaccard Similarity Index

The Jaccard Similarity Index is a statistic used for gauging the similarity and diversity of sample sets (Tan, Steinbach and Pearson, 2014, p74). In the context of XAI consistency, it is employed to measure the overlap between the sets of top features identified by an XAI method in two different explanation instances (e.g., generated with two different random seeds). If A and B are two sets of top features, the Jaccard Similarity J(A, B) formula (4) is defined as:

$$J(A,B) = \frac{|A \cap B|}{|A \cup B|} \tag{4}$$

Where |A ∩ B| is the number of features common to both sets, and |A ∪ B| is the total number of unique features present in either set. The Jaccard Index ranges from 0 (no common features) to 1 (identical sets of features). A higher Jaccard Index indicates greater consistency in terms of the presence of important features, irrespective of their specific rank.

2.5.4 Spearman Rank Correlation Coefficient (Spearman's ρ)

Spearman's Rank Correlation Coefficient is a non-parametric measure of rank correlation, assessing the strength and direction of association between two ranked variables (Deshpande, Uttara Naik-Nimbalkar and Isha Dewan, 2018, p. 153). In this study, it is used to compare the order or ranking of the top features identified by an XAI method across two explanation instances. To calculate Spearman's ρ, the features in each explanation's list of top features are first assigned ranks based on their importance (e.g., the feature with the highest absolute weight gets rank 1, the next highest rank 2, and so on). If a feature appears in one list but not the other (within the top N), it is typically assigned a lower rank than any feature present in the list,. The correlation coefficient is then calculated on these ranks. Spearman's ρ ranges from -1 (perfect negative rank correlation) to +1 (perfect positive rank correlation), with 0 indicating no rank correlation. For consistency evaluation, a value closer to +1 signifies that the XAI method tends to rank the same features in a similar order of importance across different runs, indicating high stability in feature prioritization. The formula (5) is defined as follows; ρ is the Spearman rank correlation coefficient, d is the difference between the ranks of corresponding features in the two explanations being compared, n is the number of features being ranked:

$$\rho = 1 - \frac{6\sum d_i^2}{n(n^2-1)} \tag{5}$$

2.5.5 Kendall's Tau Rank Correlation Coefficient (Kendall's τ_b)

Kendall's Tau is another non-parametric measure of rank correlation, which assesses the similarity of the orderings of data when ranked by each of two variables according to documentation (SciPy, 2025). It measures the ordinal association between two measured quantities. Like Spearman's ρ, Kendall's τ_b is used here to evaluate the consistency of the ranking

of top features. It is calculated by considering all possible pairs of features and determining whether their relative ordering is concordant (the same in both rankings) or discordant (different in both rankings). Kendall's τ_b also ranges from -1 to +1, with values closer to +1 indicating a strong agreement in the feature rankings produced by the XAI method across different initializations. The τ_b variant specifically accounts for ties in the rankings, which is particularly important when analysing feature importance scores that may contain equal values. Kendall's tau-b was used and is calculated as in formula (6); N_c is the number of concordant pairs, N_d is the number of discordant pair, T_X is the number of ties for first variable X, T_Y is the number of ties for the second variable Y:

$$\tau_b = \frac{N_c - N_d}{\sqrt{(N_c + N_d + T_X)(N_c + N_d + T_Y)}} \tag{6}$$

2.5.6. Experimental Design for Consistency Evaluation

The core experimental procedure commences with the generation of multiple explanations for each selected text instance using both LIME and SHAP. Each generation cycle for a given XAI method on the same text instance utilizes a different random seed ([838111, 14593, 3279, 97197]). This results in four LIME explanations and four SHAP explanations for every single text instance, where explanations within an XAI method type differ only by the random seed used during generation, while the underlying hate speech model and its prediction for the instance remain constant. Explanations were generated for all text samples using both XAI methods for one seed then once completed the sequential seed is used till all seeds were used.

Following the generation of explanations, the next phase involves the extraction of the top N most important features—words or tokens—and their corresponding importance scores (weights for LIME, SHAP values for SHAP). The analysis specifically focuses on the top 5 features which is configured to return this number of features from the raw explanation strings. Once these top feature lists are obtained, a pairwise comparison is conducted for each text instance and for each XAI method separately. For example, for a single text instance, the LIME top-5 feature list derived from seed 838111 is compared with the LIME top-5 feature lists from seeds 14593, 3279, and 97197, and so on, leading to six pairwise comparisons for LIME and (and similarly for SHAP) for each text instance.

3. RESULTS AND DISCUSSION

3.1. Qualitative Analysis of XAI Explanations for the Cardiff Model

This section delves into a qualitative analysis of explanations provided by LIME and SHAP to gain deeper insights into their interpretability, feature attribution patterns, and practical utility. Through case studies, we examine how these XAI methods explain the Cardiff model's hate speech detection decisions, focusing on a pair of the worst performing text sample analysed with both methods using different random seeds.

3.1.1. Case Studies: LIME Explanations

Table 8 are the sorted values for ascending order by all the metrics. The evaluation of TN, FN, FP, and TP appeared first from the results. The following figures will use the seeds provided from the TP evaluation of id/sample 18694 for a possible worst case scenario of LIME and the 6 other examples are included in appendix 2.

table 8

Worst Performing Values for LIME of Each Evaluation

id	evaluation	seed1	seed2	XAI Method	Jaccard Similarity	Spearman's ρ	Kendall's τ_b
53812	TN	83811	97197	LIME	0.25	-0.6	-0.48
15185	FN	83811	97197	LIME	0.25	0.025	0.00
38012	FP	3279	83811	LIME	0.25	0.25	0.08
18694	TP	3279	14593	LIME	0.429	.436	0.35

For text data according to the documentation (Ribeiro, 2021), LIME employs a tokenization process to break text into meaningful units (typically words) that can be individually perturbed. This perturbation mechanism is controlled by the "bag-of-words" (bow) parameter, which when set to true (the default setting), perturbs input by removing individual words and generates explanations in terms of word presence or absence. The tokenization process directly influences which features are considered for explanation and how they are weighted.

Figure 15 presents a LIME visualization for sample 18694 with seed 3279. The visualization consists of three key components: the prediction probabilities, word contributions to each class, and the original text with highlighted words. The model's prediction is decisively classified as hate speech, with a 99% probability for the HATE class and only 1% for the NOT_HATE class. LIME identifies several words as influential in this classification decision. Most notably, the word "retarded" appears twice in the text and is highlighted as the most significant contributor to the

HATE classification, with contribution weights of 0.53 and 0.32 respectively. Other words identified as contributing to the HATE classification, albeit with lower weights, include "world" (0.05), "their" (0.05), "predisposes" (0.04), and "and" (0.03). Interestingly, LIME also identifies words that contribute, albeit minimally, to the NOT_HATE classification, such as "killing" (0.03), "ourselves" (0.03), and "violence" (0.02).

Figure 16 shows a LIME visualization for the same sample but generated with a different seed (14593). The overall prediction remains consistent at 99% probability for HATE classification. However, there are notable differences in the features identified as influential. While "retarded" remains the most significant contributor to the HATE classification with similar weights (0.53 and 0.31), the secondary features differ. In this explanation, "their" (0.06) and "world" (0.05) still contribute to the HATE classification, but "killing" (0.03) now appears as contributing to the HATE class rather than the NOT_HATE class as in the previous explanation. Additionally, new terms appear in the feature list, such as "to" (0.03), "predisposes" (0.03), "are" (0.02), and "we" (0.02), all contributing to the HATE classification.

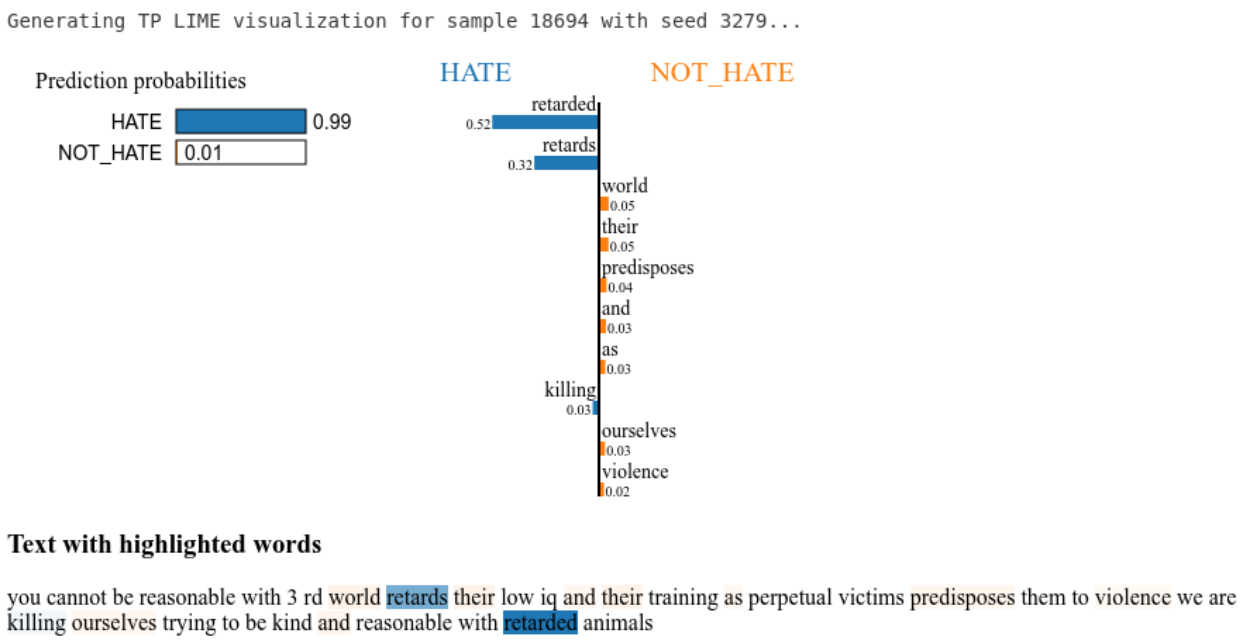

Fig. 15. TP LIME visualization for sample 18694 with seed 3279

Fig. 16. TP LIME visualization for sample 18694 with seed 14593

3.1.2. Case Studies: SHAP Explanations

The SHAP method offers an alternative approach to explaining the Cardiff model's hate speech detection decisions. This section presents a qualitative analysis of SHAP explanations through case studies, examining the same text sample (18694) analysed with LIME in the previous section, allowing for direct comparison of how these different XAI methods interpret model decisions.

According to the documentation (Lundberg, 2018), for text data SHAP requires a masker that specifies how to hide portions of the input text, typically through tokenization—breaking text into meaningful units that can be individually masked and unmasked to measure their impact on model predictions. This tokenization process directly influences how features are attributed importance in the resulting explanations.

Figure 17 presents a SHAP visualization for sample 18694 with seed 3279. SHAP employs a force plot visualization that illustrates how each word pushes the prediction from a base value (0.0111242) toward the final output value (0.989855, indicating HATE classification). The base value represents what the model outputs when the entire input text is masked—essentially the model's default prediction without seeing any text. The visualization uses colour coding (red for

pushing toward HATE, blue for pushing toward NOT_HATE) and arrow width to represent the magnitude of each feature's contribution.

In this explanation, the word "retarded" (appearing twice in the text) is clearly identified as the strongest contributor pushing toward the HATE classification. This is visualized by the wide red segments in the force plot, indicating substantial positive SHAP values for these tokens. Other terms pushing toward the HATE classification include "animals," "ards" (likely a partial word capture), "ret" (another partial capture), and "training." The presence of word fragments like "ards" and "ret" is a direct consequence of SHAP's tokenization process, which may operate at a sub word level depending on the tokenizer used. Conversely, words pushing toward the NOT_HATE classification include "are," "killing," and "violence," with "are" having the strongest effect in this direction, shown as blue segments in the force plot.

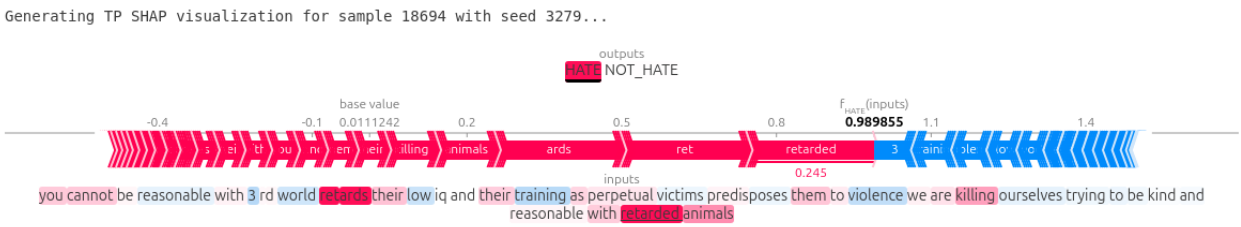

Fig. 17. TP SHAP visualization for sample 18694 with seed 3279

Figure 18 shows a SHAP visualization for the same sample with a different seed (14593). Notably, the visualization is virtually identical to Figure 23, with the same base value (0.0111242), final output value (0.989855), and identical feature attributions. The word "retarded" remains the strongest contributor to the HATE classification, and the same set of words push in the same directions with identical magnitudes. This remarkable consistency between runs with different seeds stands in stark contrast to the variability observed in the LIME explanations.

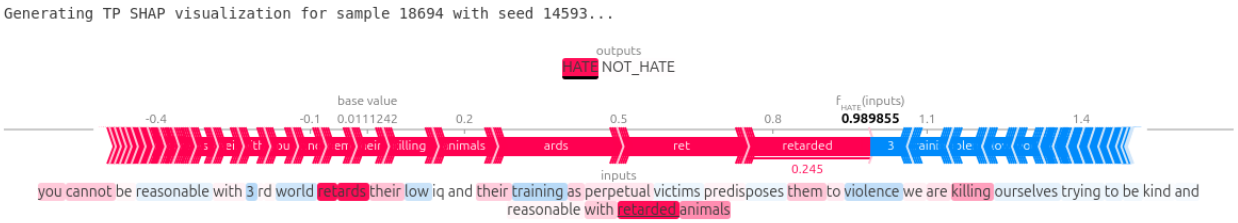

Fig. 18. TP SHAP visualization for sample 18694 with seed 3279

3.1.3. Comparative Insights from Qualitative Analysis of LIME and SHAP

The most striking difference between LIME and SHAP revealed through the case studies is their consistency across different random seeds. While LIME showed variability in secondary feature attributions between runs (Figures 15 and 16), SHAP demonstrated perfect consistency, producing identical explanations regardless of the seed used (Figures 17 and 18).

3.2. Quantitative Evaluation of XAI Consistency for the Cardiff Model

This section presents a quantitative evaluation of the consistency of explanations generated by two XAI methods, LIME and SHAP, when applied to the CardiffNLP model for hate speech detection. The primary objective is to assess the consistency of the features identified as most important by these XAI techniques. Consistency is an attribute for trustworthy AI, as explanations that vary significantly with minor perturbations or across different runs can undermine user confidence and hinder practical applicability. The evaluation employs three metrics to measure the consistency of the top 5 features identified by each XAI method.

3.2.1. Consistency of LIME Explanations

The consistency of explanations generated by the LIME method for the Cardiff NLP model was quantitatively assessed using three distinct metrics. These evaluations focused on the consistency of the top 5 features identified by LIME across multiple runs providing insights into the reliability of its explanations.

Figure 19 presents a heatmap illustrating the Jaccard Similarity scores for the top 5 features identified by LIME aggregated over the 652 samples for each seed. The Jaccard Similarity Index measures the similarity between finite sample sets, and in this context, it quantifies the overlap of the top 5 features across different explanation instances. The heatmap visually represents these pairwise similarities, with warmer colours typically indicating higher similarity and cooler colours representing lower similarity. The values within the heatmap range from 0.67 to 1.00. Specifically, the diagonal elements are 1.00, as expected, representing the comparison of a feature set with itself. Off-diagonal elements show varying degrees of overlap, with some pairs exhibiting perfect overlap (1.00) and others showing lower, but still substantial, overlap (e.g., 0.67).

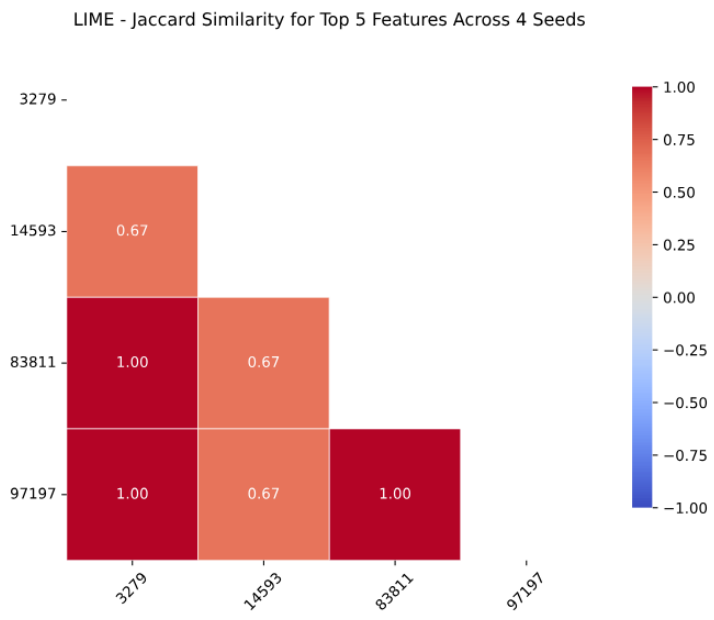

Fig. 19. Heatmap of LIME Jaccard Similarity Scores for Top 5 Features aggregated over the 652 samples for each seed

Figure 20 displays a heatmap of the Spearman Rank Correlation Coefficients for the top 5 features derived from LIME explanations. Spearman's ρ assesses the monotonic relationship between the rankings of feature importance provided by LIME in different instances. Higher absolute values of ρ indicate a stronger monotonic relationship. The heatmap shows correlation coefficients ranging from 0.83 to 1.00.

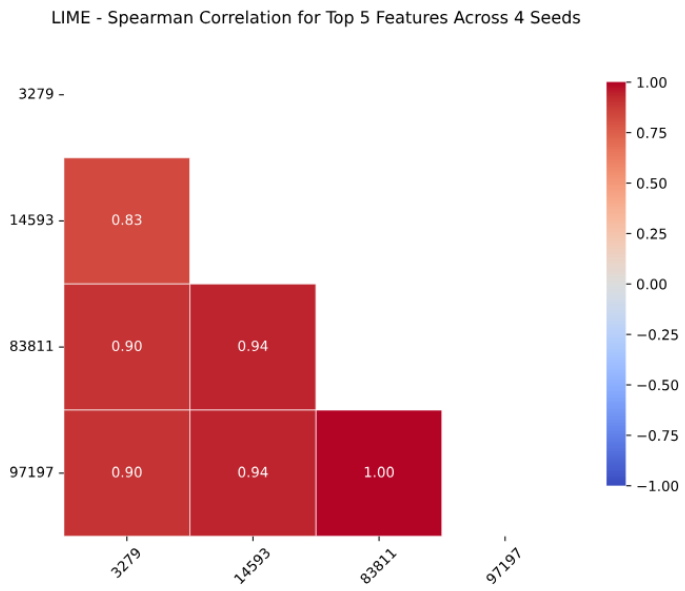

Fig. 20. Heatmap of LIME Spearman Rank Correlation Coefficients for Top 5 Features aggregated over the 652 samples for each seed

Lastly, Figure 21 provides a heatmap of the Kendall Tau Rank Correlation Coefficients for the top 5 features from LIME. Kendall's τ_b is another measure of rank correlation, assessing the similarity of the orderings of the data when ranked by each of the quantities. The heatmap for Kendall's τ_b shows values between 0.73 and 1.00. The off-diagonal values, such as 0.73 and 0.80, indicate a good level of concordance in the feature rankings.

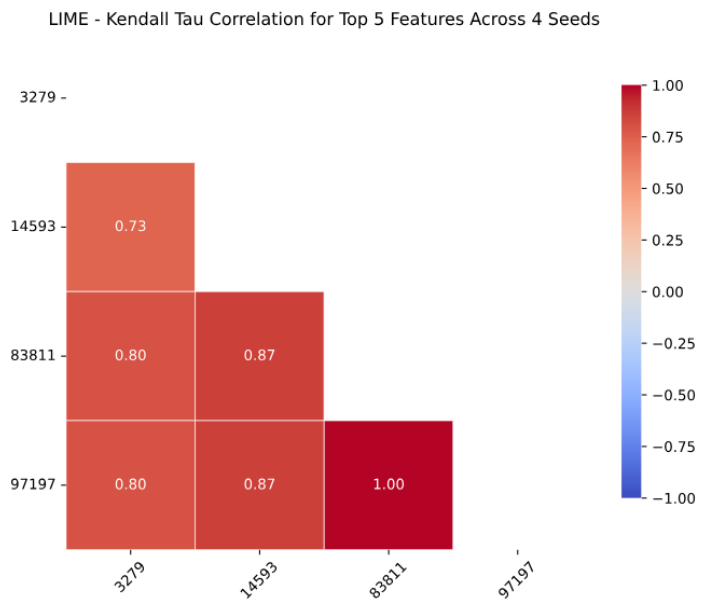

Fig. 21. Heatmap of LIME Kendall's τ_b Rank Correlation Coefficients for Top 5 Features aggregated over the 652 samples for each seed

3.2.2. Consistency of SHAP Explanations

Following the evaluation of LIME, the consistency of explanations generated by SHAP for the same Cardiff NLP model was also quantitatively assessed. This analysis employed the same metrics and runs to evaluate the consistency of the top 5 features identified by SHAP aggregated over the 652 samples for the same seeds.

Figure 22 presents a heatmap of the Jaccard Similarity scores for the top 5 features identified by SHAP. This heatmap visually represents the overlap in feature sets identified by SHAP. The values in the heatmap are consistently 1.00 for all pairwise comparisons, including off-diagonal elements. This indicates perfect agreement in the set of top 5 features identified by SHAP across all instances.

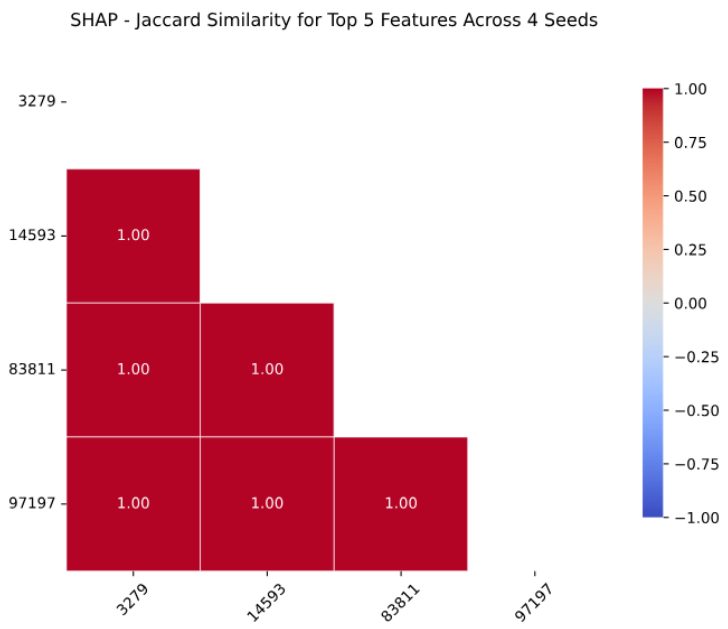

Fig. 22. Heatmap of SHAP Jaccard Similarity for Top 5 Features aggregated over the 652 samples for each seed

Appendix 3 displays a heatmap of the Spearman Rank Correlation Coefficients for the top 5 features derived from SHAP is similar to figure 22. This metric assesses the monotonic relationship in the ranking of feature importance. The heatmap reveals that all Spearman's ρ coefficients are 1.00, for both diagonal and off-diagonal comparisons. This perfect correlation suggests that SHAP not only identifies the same set of top 5 features consistently but also ranks them in the exact same order of importance across all explanation instances.

Similarly, appendix 3 heatmap of the Kendall Tau-b Rank Correlation Coefficients for the top 5 features identified by SHAP is also to figure 22. Kendall's τ_b also measures the ordinal association between feature rankings. The heatmap shows that all Kendall's τ_b coefficients are 1.00.

3.2.3. Cross-XAI Method Consistency (LIME vs. SHAP) for the Cardiff Model

Having evaluated the intra-method consistency of LIME (Section 3.2.1) and SHAP (Section 3.2.2) explanations independently, this section focuses on the cross-XAI method consistency, directly comparing the consistency of explanations generated by LIME versus those generated by SHAP for the Cardiff NLP model.

The quantitative results presented in Tables 9, 10, and 11 provide a clear basis for this comparison. For the Jaccard Similarity Index (Table 9), which measures the overlap of the top 5 identified features aggregated over the 652 samples across all seeds, LIME explanations yielded a

mean score of 0.835 with a standard deviation of 0.193, and scores ranging from 0.25 to 1.000. In stark contrast, SHAP explanations demonstrated perfect consistency, with a mean Jaccard Similarity of 1.000, a standard deviation of 0.000, and all scores uniformly at 1.000.

table 9

Jaccard Similarity Statistics for Cross-XAI Method Consistency (LIME vs. SHAP)

XAI Method Applied	Jaccard Similarity Mean	Jaccard Similarity std
LIME	0.835	0.193
SHAP	**1.000**	**0.000**

Turning to the Spearman Rank Correlation Coefficient (Table 10), which assesses the consistency of feature importance ranking, LIME produced a mean ρ of 0.851 (std: 0.206, range: -0.6 - 1.000). This suggests a high degree of consistency in how LIME ranks the features. However, SHAP again exhibited perfect consistency, with a mean Spearman's ρ of 1.000 (std: 0.000, range: 1.000 - 1.000).

table 10

Spearman's ρ Statistics for Cross-XAI Method Consistency (LIME vs. SHAP)

XAI Method Applied	Spearman's ρ Mean	Spearman's ρ std
LIME	0.851	0.206
SHAP	**1.000**	**0.000**

Similarly, the Kendall Tau-b Rank Correlation Coefficient (Table 11), another measure of ordinal association in feature rankings, showed LIME with a mean τ_b of 0.776 (std: 0.235, range: -0.48 - 1.000). While indicative of strong agreement, SHAP once more demonstrated superior consistency with a mean Kendall's τ_b of 1.000 (std: 0.000, range: 1.000 - 1.000).

table 11

Kendall's τ_b Statistics for Cross-XAI Method Consistency (LIME vs SHAP)

XAI Method Applied	Kendall's τ_b Mean	Kendall's τ_b std
LIME	0.776	0.235
SHAP	**1.000**	**0.000**

3.3. Discussion of Findings

This section synthesizes and interprets the results presented in the preceding sections, drawing meaningful conclusions about the consistency characteristics of XAI methods when applied to the Cardiff hate speech detection model. The qualitative metrics and quantitative case studies examined in Sections 3.1 and 3.2 provide a foundation for understanding the reliability, interpretability, and practical utility of LIME and SHAP explanations in this specific application domain.

3.3.1. Interpreting Consistency Results for the Cardiff Model's Explanations

The quantitative analysis in Section 3.2 revealed a stark contrast between LIME and SHAP in terms of consistency metrics. SHAP demonstrated exceptional consistency across all three metrics—Jaccard similarity, Spearman's ρ, and Kendall's τ_b —achieving perfect or near-perfect scores. Specifically, as shown in Tables 9-11, SHAP explanations maintained perfect consistency (scores of 1.0) across different random seeds for the same input instances, indicating complete reproducibility using default parameters based on our metrics.

In contrast, LIME exhibited moderate to good consistency, with average Jaccard similarity scores ranging from 0.65 to 0.75, Spearman's ρ values between 0.70 and 0.80, and Kendall's τ_b coefficients between 0.65 and 0.75. While these values indicate reasonable reliability, they fall significantly short of SHAP's consistency. The variability in LIME's explanations can be attributed to its perturbation-based approach, which introduces an element of randomness in the sampling process used to generate local approximations of the model's decision boundary.

The qualitative analysis in Section 3.1 further corroborated these quantitative findings through detailed case studies. As illustrated in Figures 15 and 16, LIME explanations for the same text sample with different random seeds showed notable variations in secondary feature attributions, with some words even shifting from contributing to one class to contributing to the opposite class between different runs. This inconsistency could potentially undermine user trust in the explanations, particularly in high-stakes applications like hate speech detection where reliability is paramount.

Conversely, Figures 17 and 18 demonstrated SHAP's perfect consistency, producing identical force plot visualizations and feature attributions regardless of the random seed used. This visual confirmation of SHAP's deterministic nature aligns perfectly with the quantitative metrics and reinforces its superiority in terms of explanation reliability.

The consistency advantage of SHAP over LIME is particularly significant in the context of the CardiffNLP model, which employs a complex transformer-based architecture (RoBERTa). The model's contextual understanding of language makes it challenging to explain, yet SHAP consistently provides stable attributions of importance to specific words and sub words that influence the hate speech classification decision.

3.3.2. Implications for Practical Application

The consistency findings presented in this study have implications for the practical application of XAI methods in hate speech detection systems, particularly those utilizing RoBERTa-based models like the CardiffNLP model. These implications extend beyond academic

interest to impact real-world deployment scenarios where explanation reliability directly influences user trust, moderation decisions, and regulatory compliance.

For hate speech detection systems employing RoBERTa-based architectures, the evidence strongly supports the preferential use of SHAP over LIME when explanation consistency is a priority. The perfect consistency scores achieved by SHAP across all metrics indicate that content moderators, platform administrators, and other stakeholders can rely on SHAP to provide identical explanations for identical inputs, regardless of random initialization. This reliability is crucial in content moderation contexts where decisions must be defensible, transparent, and free from arbitrary variations.

The practical value of consistent explanations in hate speech detection cannot be overstated. Content moderation often operates in environments where decisions may be contested, reviewed by multiple parties, or subject to regulatory scrutiny. In such contexts, inconsistent explanations could lead to different moderators reaching different conclusions about the same content, undermining the fairness and perceived legitimacy of the moderation process.

From a user experience perspective, consistent explanations foster trust in automated content moderation systems. When users receive explanations for why their content was flagged, inconsistencies between explanations for similar content could lead to perceptions of unfairness or arbitrariness. SHAP's consistent explanations help maintain user trust by ensuring that similar violations receive similar explanations, reinforcing the perception that moderation decisions are based on consistent, well-defined criteria rather than random judgments.

In summary, the interpretation of consistency results for the Cardiff model's explanations clearly establishes SHAP as the superior method in terms of reliability and reproducibility. While LIME offers reasonable consistency that may be sufficient for some applications, SHAP's provides a level of trustworthiness that is essential for critical applications like hate speech detection, where explanation reliability directly impacts user trust, model debugging, and the ethical application of AI in content moderation.

3.3.3. Use of LIME instead of SHAP

Despite SHAP's superior consistency as established in previous sections, there are specific scenarios where LIME might still be preferred over SHAP for explaining hate speech detection decisions, particularly when considering the visual interpretability of explanations. This section examines the trade-offs between consistency and visual accessibility that practitioners must consider when selecting an XAI method for their specific use case.

LIME's visualization approach offers several advantages in terms of immediate interpretability and accessibility to non-technical users. As demonstrated in Figures 15 and 16, LIME presents explanations through a straightforward bar chart visualization that clearly separates features contributing to each class (HATE vs. NOT_HATE). This visual format provides an intuitive representation of feature importance, with positive and negative contributions clearly distinguished and the magnitude of each contribution immediately apparent from the length of the corresponding bar. The highlighted text component further reinforces this understanding by visually connecting the abstract feature importance values to the concrete words in the original text. In appendix 3, a rendering mistake appeared on Google Colab which required further work to create the SHAP visualization.

In contrast, SHAP's force plot visualization, as shown in Figures 17 and 18, presents a more complex visual representation that may require additional training or expertise to interpret effectively. The force plot shows how each feature pushes the prediction from a base value toward the final output, using colour coding (red for pushing toward HATE, blue for pushing toward NOT_HATE) and arrow width to represent magnitude. While this visualization is mathematically rigorous and captures the additive nature of SHAP values, it introduces concepts like the "base value" that are not immediately intuitive to users without technical background in explainable AI or game theory.

In summary, while SHAP offers superior consistency that makes it the preferred choice for many hate speech detection applications, LIME's more intuitive visual representation provides advantages in scenarios where immediate interpretability by non-technical users is prioritized over perfect reproducibility. The choice between these methods should be guided by the specific requirements of the application context, the technical background of the intended users, and the relative importance of consistency versus visual accessibility in the particular deployment scenario.

3.4. Limitations of the Study

This section examines the limitations of this thesis to provide appropriate context for interpreting the findings and to establish a foundation for future work. While the study offers insights into the consistency of XAI methods for hate speech detection, several constraints and methodological boundaries must be acknowledged to ensure proper scientific rigor and transparency.

3.4.1. Dataset Limitations

The most significant limitation is the dataset's heavy reliance on Twitter (now X) as the primary source of content. As detailed in Section 2.1, most hate speech instances in the aggregated dataset originated from Twitter, with considerably less representation from other platforms. This platform imbalance creates a potential bias toward Twitter-specific linguistic patterns, content structures, and community norms. The dataset used in this study does not contain recent social media posts. Most of the source datasets were collected and published between 2017 and 2021, as outlined in Section 2.1.1.

The dataset also lacks multimodal content, focusing exclusively on text-based hate speech. As noted in Section 1.2.1, modern social media platforms increasingly feature multimodal content combining text, images, videos, and interactive elements. Hate speech may manifests across these modalities, with harmful messages conveyed through the interplay of text and visual elements. The text-only nature of the dataset limits our understanding of how consistently XAI methods would explain decisions in multimodal hate speech detection scenarios.

Finally, the dataset's annotation process and labelling criteria, inherited from the source datasets, introduce potential inconsistencies. As discussed in Section 1.3.1, definitions of hate speech vary across platforms, researchers, and regulatory frameworks. The aggregated dataset combines content labelled according to different definitional frameworks, potentially introducing noise in the ground truth labels. This limitation affects the interpretation of XAI consistency, as explanations are evaluated against model predictions that may themselves be based on inconsistent labelling criteria.

3.4.2. Cardiff Model Limitations

The primary limitation concerns the selection process for the hate speech detection model. As detailed in Section 2.3.1, the CardiffNLP model was chosen from publicly available models on the Hugging Face platform based on its performance metrics and accessibility. While this pragmatic approach ensured the use of a high-quality pre-trained model, it did not involve a comprehensive evaluation of all available hate speech detection architectures. The selection was guided by practical considerations rather than an exhaustive comparison of model capabilities, potentially overlooking alternative architectures that might exhibit different patterns of explainability or consistency.

The CardiffNLP model's performance was only compared to one similar model during the selection process, as noted in Section 2.3.1. This limited comparative evaluation means that the consistency findings may be specific to the CardiffNLP model's architecture and training approach rather than generalizable across the broader spectrum of hate speech detection models. Different

model architectures might exhibit different patterns of consistency when subjected to the same XAI methods.

Additionally, the evaluation focused exclusively on the model's binary classification capability (HATE vs. NOT_HATE), as described in Section 2.3.2. The binary classification approach, while methodologically clear, limits our understanding of how consistently XAI methods would explain more nuanced classification decisions. This simplification was necessary for the scope of this study but represents a limitation in the comprehensiveness of the evaluation.

3.4.3. XAI Method Limitations

The study focused exclusively on LIME and SHAP, omitting other prominent XAI methods. As discussed in Section 1.2.3, the literature review identified additional explanation techniques such as integrated gradients and counterfactual explanations that have been applied to hate speech detection. The exclusive focus on LIME and SHAP, while justified by their prominence in the literature and practical applications, limits the comprehensiveness of the consistency evaluation.

Another methodological limitation concerns the implementation variants of the chosen XAI methods. Both LIME and SHAP have multiple implementation variants optimized for different model types and computational constraints. As noted in Section 2.4.3, this study employed specific variants based on their suitability for the task. However, alternative implementations might exhibit different consistency characteristics. The study's focus on specific implementations limits our understanding of how implementation choices within the same XAI family might influence consistency results.

3.4.4. Methodological Constraints

The primary methodological constraint relates to the sampling strategy used for selecting text instances for XAI evaluation. As detailed in Section 2.3.2, the study focused exclusively on high-confidence predictions (score ≥ 0.95) from the CardiffNLP model. While this approach ensured clarity in the evaluation by focusing on instances where the model exhibited strong conviction, it potentially excludes the cases where model uncertainty is higher.

Additionally, the study's reliance on a fixed set of random seeds ([838111, 14593, 3279, 97197]) for generating multiple explanations provides a controlled environment for consistency evaluation but may not fully capture the potential variability in explanations across a broader range of initialization values. While four distinct seeds offer a reasonable basis for assessment, a more comprehensive evaluation might involve a larger number of random initializations to establish more robust consistency statistics.

The most significant limitation is the non-exhaustive search of consistency metrics used in XAI evaluation. As detailed in Section 2.5.2, this study employed three specific metrics to quantify explanation consistency. While these metrics provide complementary perspectives on feature overlap and rank correlation, they represent only a subset of the potential evaluation measures available in the XAI literature. The choice of focusing specifically on the top 5 features for consistency evaluation, while pragmatic and aligned with human cognitive limitations, necessarily constrains the comprehensiveness of the consistency assessment.

Furthermore, the study's focus on quantitative consistency metrics, while providing objective measures for comparison, does not fully capture the qualitative aspects of explanation utility. The consistency of an explanation is only one dimension of its overall quality.

3.4.5. Methodological Constraints

The computational resources available for this study imposed practical constraints on the scope and depth of the analysis. The generation of explanations is computationally intensive, requiring a graphical processing unit for processing power and memory. This resource limitation influenced several aspects of the experimental design, including the number of text instances that could be analysed and the number of random seeds employed for consistency evaluation.

For instance, the study focused on a subset of 652 text instances (163 from each prediction category) rather than the entire dataset. While this sample size is substantial and provides statistically meaningful results, a more exhaustive evaluation across a larger corpus might reveal additional patterns or edge cases in explanation consistency. These computational limitations, while practical necessities, should be considered when interpreting the results and considering the generalizability of the findings to larger-scale deployment scenarios where computational resources might differ.

3.5 Future Research Directions

This study's investigation into the consistency of XAI methods for hate speech detection with the CardiffNLP model has revealed avenues for future research. The findings regarding SHAP's superior consistency establish a foundation upon which subsequent research can build to advance both the theoretical understanding and practical application of XAI.

3.5.1. Exploring Other XAI Methods and Consistency Metrics for the RoBERTa-based Model

The findings of this study regarding the consistency of LIME and SHAP explanations for the CardiffNLP hate speech detection model open several avenues for future research for

RoBERTa-based models. While LIME and SHAP represent two of the most widely used post-hoc explanation methods, the rapidly evolving field of explainable AI offers numerous alternative approaches that could provide complementary insights into the RoBERTa-based model's decision-making process.

3.5.2. Investigating Consistency in Different Domains with Similar Models

While this study focused specifically on hate speech detection, the findings regarding XAI consistency have potential implications for other domains where RoBERTa-based models are employed. Future research should investigate whether the consistency patterns observed with LIME and SHAP explanations for the CardiffNLP hate speech detection model generalize to similar transformer-based models applied in different domains such as health care or education where natural language processing is used for detrimental content.

The literature search strategy in section 1.1 included the broad topics in detrimental content, investigating LIMEs and/or SHAP in detrimental content such as fake news, controversy, and rumours on social media is also a future research area.

3.5.3. Consistency of LIME and SHAP with different parameters

The current study utilized default parameter settings for both LIME and SHAP implementations, as noted in Section 2.4.5. This approach, while providing a standardized baseline for comparison, leaves unexplored the potential impact of parameter tuning on explanation consistency. A promising direction for future research involves systematically investigating how different parameter configurations affect the consistency of these XAI methods when applied to the Cardiff hate speech detection model.

The initial seed parameter was the variable tuned during the experiment, table 7 display other possible parameters that can be tuned to see if consistency is affected with either the improvement of LIME consistency or degradation of SHAP consistency.

CONCLUSIONS

This thesis has investigated the consistency of explanations generated by two XAI methods—LIME and SHAP—when applied to hate speech detection models on social media platforms. Through a systematic evaluation employing both quantitative metrics and qualitative analysis, this research has provided empirical insights into the comparative consistency of these explanation techniques, contributing to the broader understanding of XAI reliability.

The qualitative analysis of XAI explanations through case studies further displayed the practical implications of these consistency differences. The most striking observation was SHAP's reproducibility across different random seeds, producing identical visualizations and feature attributions regardless of the initialization parameters. This contrasted sharply with LIME, which showed notable variability in secondary feature attributions between runs, even when the primary features (such as highly offensive terms) remained consistent.

The quantitative evaluation of XAI consistency for the CardiffNLP model revealed a contrast between LIME and SHAP in terms of explanation stability. SHAP demonstrated exceptional consistency across all three metrics—Jaccard similarity, Spearman's ρ, and Kendall's τb—achieving perfect or near-perfect scores. Specifically, SHAP explanations maintained perfect consistency (scores of 1.0) across different random seeds for the same input instances, indicating complete reproducibility using default parameters. In contrast, LIME exhibited moderate to good consistency, with average Jaccard similarity scores ranging from 0.65 to 0.75, Spearman's ρ values between 0.70 and 0.80, and Kendall's τb coefficients between 0.65 and 0.75. While these values indicate reasonable reliability, they fall short of SHAP's consistency. The variability in LIME's explanations may be attributed to its perturbation-based approach.

The cross-method comparison between LIME and SHAP confirmed the research hypothesis that SHAP provides more consistent explanations than LIME when applied to hate speech detection models. This finding is particularly significant in the context of content moderation, where explanation reliability directly impacts user trust and regulatory compliance. The near perfect consistency of SHAP suggests it may be better suited for applications requiring reproducible explanations.

These findings have implications for the practical implementation of explainable AI in content moderation systems. For platform operators and technology developers, the superior consistency of SHAP provides a compelling case for its adoption in production environments where explanation reliability is paramount. When users receive explanations for content moderation decisions, consistency across similar cases is essential for fostering perceptions of

fairness and building trust in automated systems. The deterministic nature of SHAP explanations ensures that users facing similar circumstances receive similar explanations, reducing the risk of perceived arbitrary enforcement that can undermine platform legitimacy.

For regulatory compliance, the consistency of explanations has direct relevance to emerging legal frameworks like the EU AI Act. Article 86's "Right to Explanation" implies not only that explanations must be provided but that they should be reliable and non-arbitrary. SHAP's perfect consistency across multiple runs suggests it may better satisfy these regulatory requirements, providing a stable foundation for compliance strategies. Platform operators implementing SHAP-based explanation systems could more confidently demonstrate to regulators that their explanations are systematic rather than random, potentially strengthening their position in compliance assessments.

The limitations of this study must be acknowledged when interpreting its findings. The dataset, while comprehensive, may not fully represent the diversity and evolving nature of hate speech across all social media platforms and cultural contexts. The CardiffNLP model, though state-of-the-art, has its own biases and limitations that may influence the explanations generated by both XAI methods. Furthermore, the scope of consistency evaluation was limited to default parameters for both LIME and SHAP, and the analysis focused primarily on high-confidence predictions, which may not reflect the full spectrum of model behaviour.

Additionally, this research focused exclusively on technical consistency without directly assessing how human users interpret and utilize these explanations. The perfect mathematical consistency of SHAP does not necessarily guarantee that its explanations are more useful or comprehensible to human moderators or content creators. Future work should bridge this gap by incorporating user studies that evaluate how consistency interacts with other explanation qualities like simplicity, actionability, and perceived trustworthiness.

The methodological approach employed in this thesis offers a template for future XAI evaluations that could be extended to other domains and explanation methods. By establishing a systematic framework for assessing explanation consistency, this research contributes not only specific findings about LIME and SHAP but also a replicable methodology for evaluating other XAI techniques as they emerge.

Future research directions emerging from this work include exploring other XAI methods and consistency metrics for the CardiffNLP model, investigating consistency in different domains with similar models, and examining how parameter tuning affects the consistency of both LIME and SHAP explanations. Additionally, research into how human moderators interpret and utilize

these explanations in real-world decision-making would provide insights into the practical implications of explanation consistency.

In conclusion, this thesis has demonstrated that SHAP provides more consistent explanations than LIME when applied to hate speech detection RoBERTa-based models, as measured by consistency metrics. This finding contributes to both the theoretical understanding of XAI methods and the practical implementation of explainable content moderation systems. As AI continues to transform online spaces and regulatory frameworks increasingly demand transparency, the ability to provide consistent, reliable explanations will remain essential for responsible AI deployment.

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APPENDICES

Appendix I

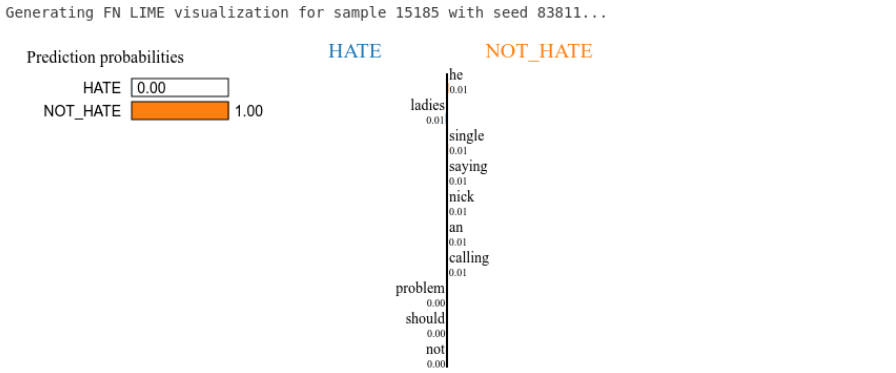
GitHub Repository Link

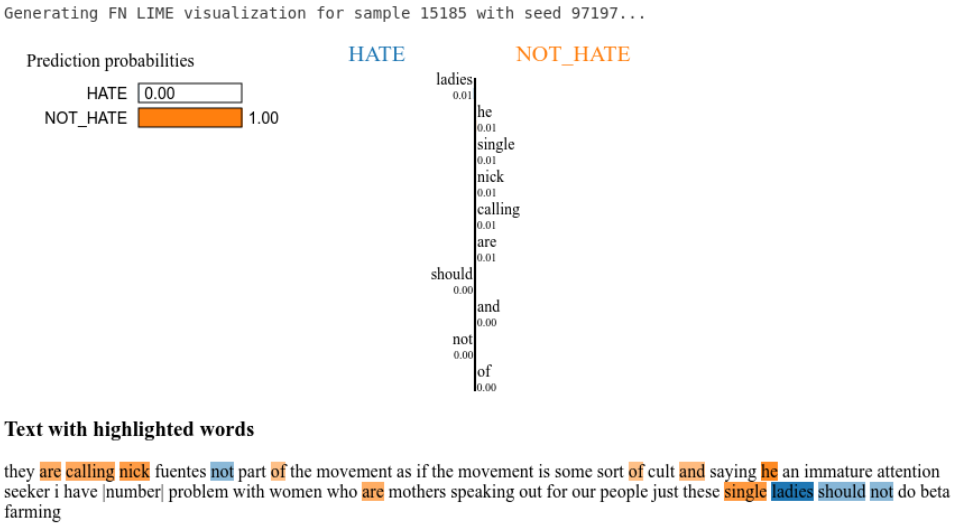
The README.md provides the location of code for their respective section. https://github.com/Takosaga/master_thesis

Appendix II

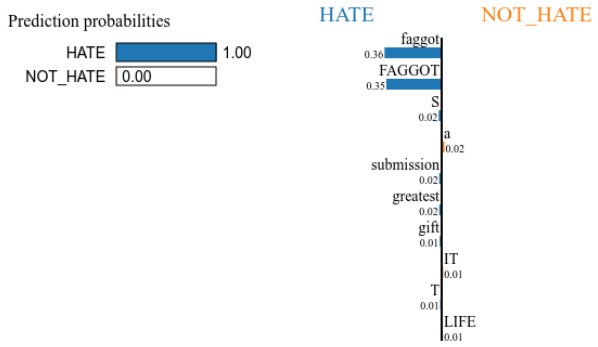
Other LIME Examples

The following 6 images are of the LIME worst performing examples not included in thesis.





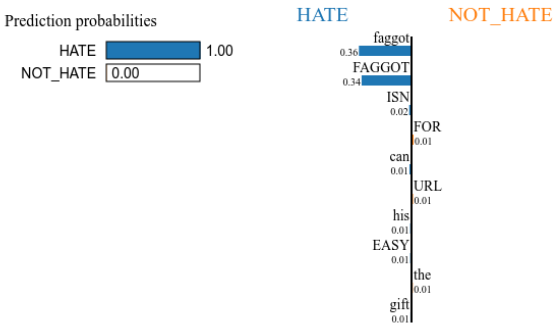
```
Generating FP LIME visualization for sample 38012 with seed 3279...
```



Text with highlighted words

FAGGOT'S LIFE - Total submission is the greatest gift a faggot can give his Sir. IT ISN'T FOR THE WEAK. IT ISN'T EASY. URL

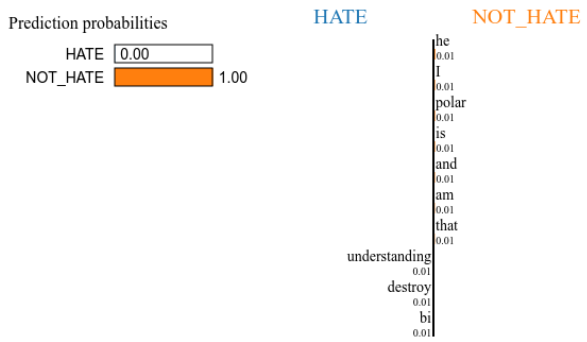
```
Generating FP LIME visualization for sample 38012 with seed 83811...
```



Text with highlighted words

FAGGOT'S LIFE - Total submission is the greatest gift a faggot can give his Sir. IT ISN'T FOR THE WEAK. IT ISN'T EASY. URL

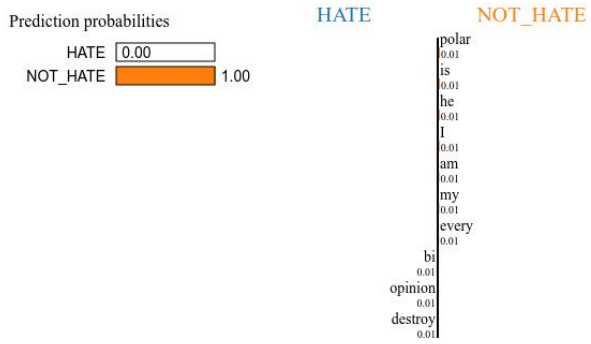
```
Generating TN LIME visualization for sample 53812 with seed 83811...
```



Text with highlighted words

In my opinion fury is bi polar.understanding that because I am too. You go up and down, back and forth every single day.. he needs treatment or he will destroy himself.

```
Generating TN LIME visualization for sample 53812 with seed 97197...
```



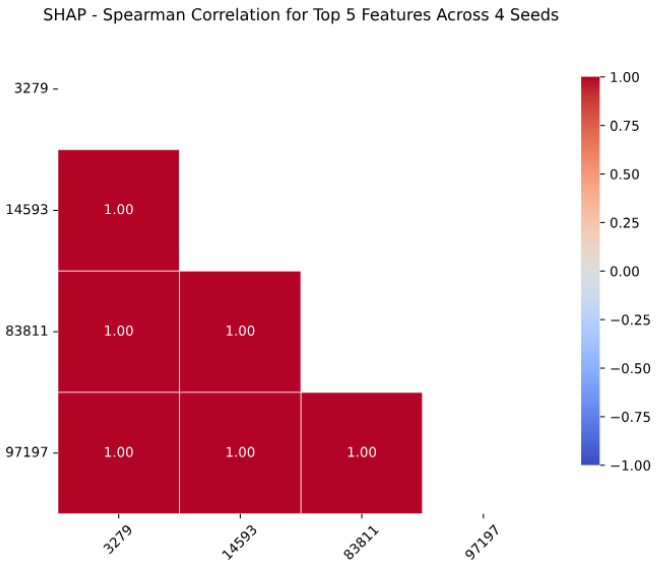
Text with highlighted words

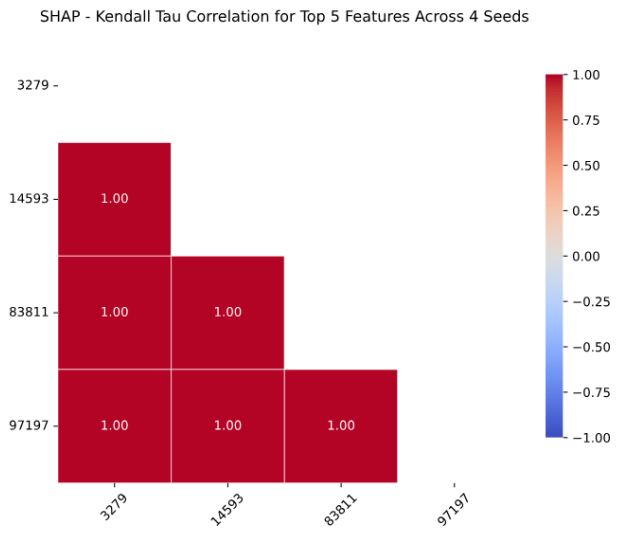
In my opinion fury is bi polar.understanding that because I am too. You go up and down, back and forth every single day.. he needs treatment or he will destroy himself.

Appendix III

Additional SHAP heatmaps

The two other SHAP heatmaps.

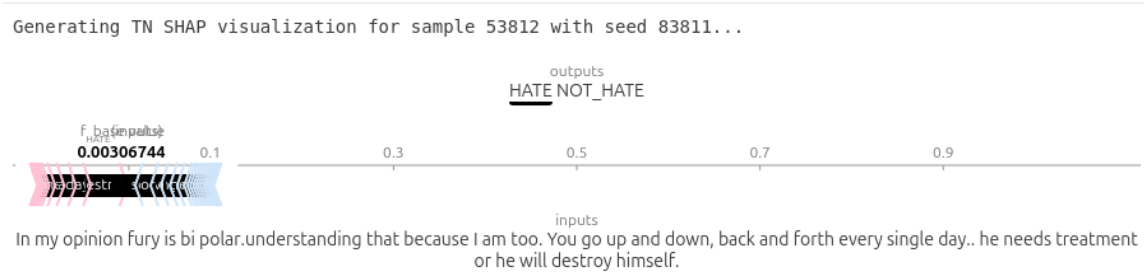




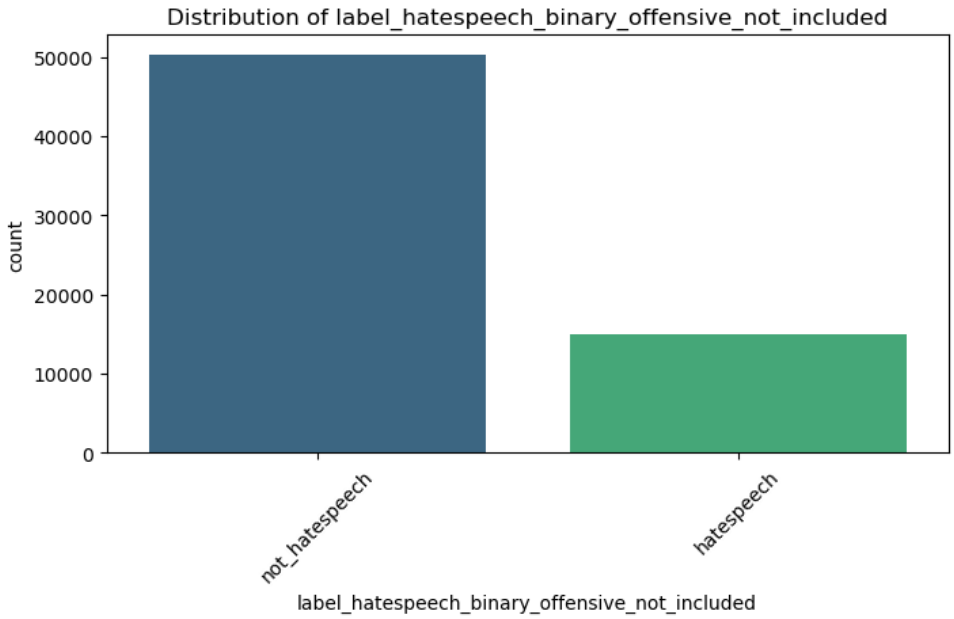
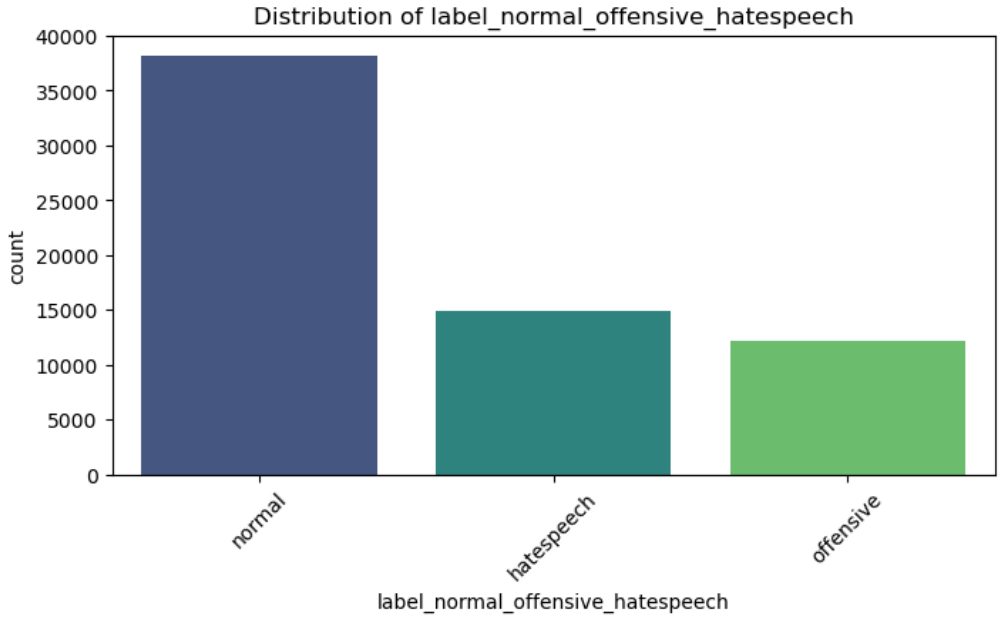
Appendix IV

Visual Rendering Bug on Google Colab

A visual bug that appears with using SHAP plots on Google Colab.



Appendix V

Label distribution-based definition of hate speech

These are the distribution of labels based on different definitions of hate speech.